

BREAST CANCER ULTRASOUND CLASSIFICATION USING MACHINE LEARNING

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Abstract

Breast ultrasound is a useful adjunct to mammography, but its interpretation is operator dependent and image appearance is affected by speckle, acquisition settings, and lesion context. This dissertation develops a reproducible research prototype for binary classification of breast-ultrasound images using convolutional neural networks. The software integrates contrast-limited adaptive histogram equalisation (CLAHE), aspect-ratio-preserving square padding, transfer-learning backbones (EfficientNet-B0 and ResNet-50), and a custom convolutional baseline. It also provides a FastAPI research demonstration for controlled image inference and visual inspection.

The principal contribution is methodological rather than clinical: a single preprocessing implementation is shared by training, evaluation, command-line inference, and the web service; paired lesion masks are recognised under the naming conventions used by the supplied datasets; and evaluation emits machine-readable metrics, figures, and an audit trail. A read-only data audit records split sizes and flags cross-split subject identifiers before performance is interpreted.

The local OASBUD and BrEaST copies were rebuilt into subject-level partitions and pass the split audit. BUSI remains excluded because its renamed image copy does not retain enough provenance to verify patient separation. A fresh EfficientNet-B0 checkpoint evaluated on 62 held-out BrEaST and OASBUD images produced 67.74% accuracy, macro F1 0.6774, and ROC-AUC 0.7419. These values describe a local research cohort, not clinical or external validation. Historical headline figures, pseudo-labelling ablations, and noise curves contained in legacy project assets are therefore not treated as empirical findings.

The corrected workflow rebuilds the cohort at subject level, preserves source identifiers and preprocessing manifests, retrains with fixed seeds, and generates tables and figures from held-out predictions. The system remains an educational research prototype, not a diagnostic or clinical decision-support device.

Declaration

No part of this project has been submitted in support of an application for any other degree or qualification at this or any other institute of learning. Apart from those parts of the project containing citations to the work of others, this project is my own unaided work. This work has been carried out in accordance with the Manchester Metropolitan University research ethics procedures, and has received ethical approval number 92016.

Signed: Akerele David Damilola

A handwritten signature in black ink, appearing to read 'Akerele', written over a light grey rectangular background.

Date: 6 September 2026

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Abbreviations

ACR	American College of Radiology
AI	Artificial Intelligence
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BI-RADS	Breast Imaging-Reporting and Data System
BUS	Breast Ultrasound
BUSI	Breast Ultrasound Images Dataset
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNR	Contrast-to-Noise Ratio
CNN	Convolutional Neural Network
ECE	Expected Calibration Error
FPR	False Positive Rate
Grad-CAM	Gradient-weighted Class Activation Mapping
MBConv	Mobile Inverted Bottleneck Convolution
OASBUD	Open Access Ultrasound Database
ROI	Region of Interest
SE	Squeeze-and-Excitation

Chapter 1

Introduction

1.1 Background

Breast cancer is a major global health burden and was the most commonly diagnosed cancer worldwide in the 2020 GLOBOCAN estimates (Sung et al. 2021). Imaging is central to detection and assessment. Ultrasound is particularly useful as a non-ionising adjunct that can characterise breast tissue and guide procedures, but its appearance varies with acquisition settings, equipment, anatomy, and operator technique (Stavros et al. 1995; Al-Dhabyani et al. 2020). These sources of variation complicate both human interpretation and the development of machine-learning systems.

Convolutional neural networks (CNNs) can learn image representations directly from pixels, and transfer learning can make these models practical when labelled medical datasets are comparatively small (Litjens et al. 2017; He et al. 2016; Tan and Le 2019). However, strong performance on a curated image split does not by itself establish patient-independent generalisation. Views from the same lesion or subject can be highly correlated, so placing related images in different partitions can inflate an evaluation estimate. Reproducibility therefore depends on data provenance and group-aware partitioning as much as on model architecture.

This dissertation develops a research prototype for binary classification of breast-ultrasound images. The project combines configurable contrast enhancement, aspect-ratio-preserving preprocessing, three CNN backbones, machine-readable evaluation artifacts, a split-integrity audit, and a local web interface. It explicitly separates what the present software demonstrates from what would require a new, audited experiment.

1.2 Problem statement

Three connected problems motivate the work.

1. Image variability. Speckle, gain, field of view, and resolution vary between images and datasets. Preprocessing must be specified and shared across training, evaluation, and inference rather than reimplemented independently.
2. Geometric transformation. Directly resizing a rectangular image to a square uses different horizontal and vertical scale factors. This can alter apparent lesion geometry. Square padding before resizing applies one isotropic scale factor, although any effect on classification must be measured rather than assumed.
3. Evaluation validity. Image-level stratification is insufficient when subjects contribute multiple views. An evaluation is interpretable only when subject identifiers, partitions, preprocessing, seeds, checkpoints, and per-image predictions are retained.

The first audit of the supplied folders exposed the third problem directly: identifiable OASBUD and BrEaST subjects crossed partitions, while renamed BUSI files did not retain enough provenance to verify patient separation. OASBUD and BrEaST were rebuilt deterministically and now pass the subject-level audit. The resulting checkpoint provides a local patient-separated estimate for those two sources. It does not estimate performance at another hospital, in clinical practice, or on the unverified BUSI derivative.

1.3 Aim and research questions

The aim is to design and critically evaluate a reproducible software framework for breast-ultrasound image classification while maintaining a clear boundary between engineering evidence and clinical claims.

The study is organised around the following questions:

- RQ1 Can a single, testable preprocessing and inference path support training, evaluation, command-line prediction, and a web demonstration?

RQ2 How should CLAHE and aspect-ratio-preserving square padding be evaluated against simpler preprocessing choices without confounding data partitions, model budgets, or random seeds?

RQ3 What conclusions are justified by the current dataset provenance and split audit, and what additional controls are required for wider external evaluation?

RQ1 is addressed through implementation and regression testing. RQ2 defines a controlled ablation protocol; the present local cohort does not support a comparative conclusion. RQ3 is addressed through the repository audit and the evidence-status analysis in Chapter 4.

1.4 Objectives

The project objectives are to:

1. review ultrasound image formation, speckle, relevant lesion descriptors, CNN transfer learning, and reproducible evaluation;
2. provide deterministic, subject-aware data-splitting and read-only audit utilities;
3. implement CLAHE and three explicitly selectable geometric preprocessing strategies;
4. implement EfficientNet-B0, ResNet-50, and a compact custom CNN using a common binary output contract;
5. record training configuration and data-audit status in saved checkpoints;
6. generate summary metrics, per-image predictions, a confusion matrix, and an ROC curve from one evaluator;
7. expose research-only inference, comparison, and controlled noise demonstrations through a local FastAPI application; and
8. keep the dissertation, notebook, documentation, web interface, Word document, and presentation consistent with the same evidence source.

1.5 Contributions

The principal contribution is not a claim of diagnostic superiority. It is an evidence-aware implementation that makes common sources of divergence visible and testable. Specifically, the repository now provides:

- a shared preprocessing function and paired-mask discovery for supported dataset naming conventions;
- a training guard that refuses an unaudited cohort unless the user explicitly requests a provisional run;
- evaluation outputs that include audit status, preprocessing configuration, per-image probabilities, and an audit checksum;
- regression tests for preprocessing dimensions, mask discovery, data-status reporting, and the research-only API boundary; and
- synchronized written and visual artifacts that withdraw unsupported historical benchmark claims.

1.6 Scope and limitations

The software performs binary image classification using 2D B-mode breast-ultrasound images. The configured classes are benign and malignant; images stored in a normal directory are mapped to the non-malignant class, which is a modelling simplification that must be reconsidered in a definitive study. The work does not establish a diagnosis, assign BI-RADS, recommend management, or validate a medical device. Heatmaps, confidence values, morphology descriptors, and image-quality estimates are exploratory model outputs only.

No pseudo-labelling experiment, preprocessing ablation, multi-seed comparison, external-site evaluation, or prospective reader study is claimed as completed. These are appropriate extensions from the audited BrEaST and OASBUD baseline. BUSI should remain outside that work unless a separate original release can be curated with traceable identifiers.

1.7 Dissertation structure

Chapter 2 reviews the physical and computational background. Chapter 3 defines the data audit, preprocessing, models, training controls, and evaluation protocol. Chapter 4 reports the current executable result and its evidence limitations. Chapter 5 discusses validity, reproducibility, and clinical boundaries. Chapter 6 concludes and specifies the work required for a defensible final experiment.

1.8 Rationale for an evidence-aware prototype

Ultrasound is attractive for a student research project because it is comparatively accessible, portable, and capable of showing soft-tissue structure without ionising radiation. Those advantages are accompanied by a difficult image-formation problem. The appearance of a lesion is influenced by probe pressure, probe orientation, focal-zone placement, gain, depth, dynamic range, and the operator’s decision about what to include in the frame. Two images of the same lesion can therefore be visually different even when the underlying pathology has not changed. Conversely, two images from different patients can share scanner-specific or operator-specific characteristics. A model trained on a small collection may learn either useful lesion evidence or these incidental regularities, and the image alone does not reveal which has happened.

The browser demonstration is only the final step in a chain that begins with file naming and dataset curation. If that chain changes between training and demonstration, the reported result no longer describes one experiment. The project treats the data manifest, preprocessing configuration, checkpoint metadata, and evaluator output as research records because implementation shortcuts can hide the assumptions behind a reported number.

The problem is consequently methodological as well as computational. It is not enough to ask which network produces the largest score on the available folders. The more useful question is whether the score can be traced to a documented population, a defensible partition, and a fixed transformation. This changes the order of work. Dataset inspection precedes model comparison; an audit precedes a headline metric; and an interface label must remain narrower than a clinical conclusion. The dissertation adopts that order deliberately, even when it means reporting a less impressive result than the original project materials

suggested.

There is also a communication problem. Medical-image projects often use the language of diagnosis even when the underlying experiment is an image-level classification exercise. Terms such as “confidence”, “sensitivity”, and “robustness” have specific meanings, and readers may reasonably infer more than the experiment supports if those terms are presented without qualification. The report therefore uses explicit evidence labels: engineering evidence refers to a functioning pipeline, local-cohort evidence refers to the audited test partition, and clinical evidence is not claimed.

The questions are narrower than those of a clinical validation programme. They ask whether the project can produce a trustworthy experimental object and leave treatment decisions outside its scope. This framing makes the work falsifiable. If the audit fails, the result is a documented failure of the proposed evidence path. If two preprocessing choices perform similarly after a controlled comparison, the report should state that outcome instead of presenting a preferred visual transformation as an improvement.

The scope is further limited by the unit of annotation. The labels are inherited from dataset folders rather than adjudicated for this study, and the available images do not provide a complete clinical record, pathology context, acquisition protocol, or reader disagreement measure. The system cannot distinguish a technically poor image from a clinically difficult case in a validated way. Its image-quality heuristics are indicators for inspection, not exclusion rules. Similarly, a saliency map can highlight a plausible region while still being unstable under small perturbations. These limitations are properties of the available evidence rather than defects that can be hidden by a more elaborate interface.

The methodology is a specification of the study design, while the results chapter records what was actually run. The discussion explains why a moderate, auditable result is more informative than a higher historical value that cannot be reproduced. This separation matters because the project includes both research code and a user-facing demonstration. The interface shows what the software can do; the dissertation explains what the evidence means.

1.9 Researcher’s position and design choices

The project was developed as an applied machine-learning study rather than as a clinical trial. That position affects the choice of language and the order of decisions. A clinical trial would begin with a defined intended use, a target population, a prespecified endpoint, and governance approvals. This dissertation begins with an inherited folder of images and asks whether it can be turned into a reproducible experiment without hiding the weaknesses of that starting point. The resulting contribution is therefore partly technical and partly documentary.

Several design choices follow from this position. The system favours explicit configuration over convenience, retains provisional outputs instead of deleting them, and reports limitations next to results. It also keeps a local web demonstration because a working interface exposes assumptions that are easy to miss in a notebook: file-size limits, missing masks, low-confidence predictions, and the difference between a model output and a user-facing label. These choices do not make the classifier clinically reliable, but they make the research object easier to inspect.

Chapter 2

Background and Literature Review

2.1 Breast ultrasound as an imaging modality

Ultrasound forms images from the transmission and reception of acoustic waves. At a simplified level, the speed of a small-amplitude pressure wave in a medium is related to its bulk modulus K and density ρ by

$$c = \sqrt{\frac{K}{\rho}}. \quad (2.1)$$

At an interface between tissues with acoustic impedances Z_1 and Z_2 , the normal-incidence intensity reflection coefficient can be written as

$$R_I = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2. \quad (2.2)$$

Spatially varying reflection, scattering, attenuation, time-gain compensation, and scanner processing jointly determine the displayed B-mode image. The resulting pixel array is therefore not a direct tissue measurement and can vary substantially between acquisitions.

Breast ultrasound is used alongside clinical history and other imaging. Radiologists may consider features such as shape, margin, orientation, echogenicity, and posterior acoustic behaviour within the ACR BI-RADS ultrasound lexicon (Mendelson et al. 2013; Stavros et al. 1995). These descriptors provide clinical context for why geometry and local contrast matter. They do not justify inferring a BI-RADS category from this project’s classifier or from its heuristic image measurements.

2.2 Speckle and image variability

Speckle arises from constructive and destructive interference among echoes from unresolved scatterers. A frequently used simplified observation model combines signal-dependent and additive components:

$$I(x, y) = f(x, y) \eta_m(x, y) + \eta_a(x, y), \quad (2.3)$$

where f is an underlying reflectivity field, η_m is a multiplicative term, and η_a represents additive acquisition noise. This model is an approximation: displayed clinical images also contain beamforming, compression, filtering, and device-specific processing.

Traditional reduction methods include local-statistics filters and speckle-reducing anisotropic diffusion (Lee 1980; Yu and Acton 2002). More recent work evaluates learned models under perturbations to determine whether apparently strong classifiers remain stable when image quality changes (Jiang et al. 2023). Robustness must be tested on stored predictions with a specified perturbation process and seed. A visually plausible synthetic example is not by itself a performance experiment.

Filtering also changes the evidence available to a later segmentation or classification stage. Osman et al. compared filtering algorithms for breast ultrasound lesion segmentation and found that preprocessing choices can alter the resulting boundary estimate (Osman et al. 2018). Yap et al. likewise studied processed ultrasound images in relation to human perception (Yap, Edirisinghe, and Bez 2010). Together, these studies support treating enhancement as an experimental factor, not as an automatic improvement.

2.3 Geometric preprocessing

CNN batches usually require a common spatial size. If an image of height H and width W is directly resized to H' by W' , its axis-specific scale factors are

$$s_y = \frac{H'}{H}, \quad s_x = \frac{W'}{W}. \quad (2.4)$$

When $s_x \neq s_y$, the transformation is anisotropic and changes apparent aspect ratios. An alternative is to crop a defined region, pad it to a square, and then resize it. If the square side is d , both axes use the same final factor $224/d$ in this project.

This geometric property is deterministic, but its effect on classification is empirical. Cropping can discard useful context; reflection padding can introduce repeated boundary structure; and mask-guided training can differ from unmasked deployment. A fair ablation must hold the partition, architecture, optimisation budget, and random seeds constant while changing only the preprocessing strategy.

2.4 Contrast Limited Adaptive Histogram Equalisation

Global histogram equalisation uses an image-wide cumulative distribution and can over-emphasise background variation. Contrast Limited Adaptive Histogram Equalisation (CLAHE) instead applies local mappings and clips histogram mass before redistribution (Zuiderveld 1994). For a contextual region with histogram $h(k)$, a conceptual clipped histogram is

$$h_c(k) = \min(h(k), \beta) + r(k), \quad (2.5)$$

where β is the clip threshold and $r(k)$ distributes the clipped mass. Interpolation between neighbouring mappings reduces tile-boundary discontinuities.

The implementation in this repository applies OpenCV CLAHE to the luminance channel in LAB colour space, with a clip limit of 2.0 and a 16×16 tile grid. This is a documented configuration choice, not evidence that CLAHE improves discrimination. Its effect requires a controlled ablation and should include both predictive and image-quality analysis.

2.5 Convolutional representation learning

CNNs learn filters through repeated convolution, non-linearity, and aggregation. For an activation tensor $\mathbf{X}^{(l-1)}$, a layer can be represented schematically as

$$\mathbf{X}^{(l)} = \sigma(\mathbf{W}^{(l)} * \mathbf{X}^{(l-1)} + \mathbf{b}^{(l)}). \quad (2.6)$$

Medical-image datasets are often much smaller than general computer-vision corpora, so transfer learning is commonly used to initialise useful low- and mid-level features (Litjens et al. 2017).

2.5.1 ResNet-50

Residual networks learn a residual function around an identity path,

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x}, \quad (2.7)$$

which supports optimisation of deeper architectures (He et al. 2016). In this project, the original classification layer is replaced by dropout and a two-output linear layer.

2.5.2 EfficientNet-B0

EfficientNet uses coordinated scaling of network depth, width, and input resolution, with the B0 model serving as the baseline architecture (Tan and Le 2019). Its mobile inverted bottleneck blocks include squeeze-and-excitation operations that reweight channels (Hu, Shen, and Sun 2018). The project replaces the classifier with dropout and a two-output linear layer.

2.5.3 Custom CNN baseline

The custom model provides a compact, from-scratch baseline with four convolutional stages, batch normalisation, ReLU activations, max pooling, adaptive average pooling, dropout, and a two-output layer. It helps distinguish the contribution of transfer learning from that of the surrounding pipeline, but a fair comparison requires equivalent split integrity and a predeclared tuning budget.

2.6 Outputs, confidence, and explanation

The two logits are converted to a mutually exclusive probability vector using softmax:

$$p(y = c \mid \mathbf{x}) = \frac{\exp(z_c)}{\sum_{j=1}^K \exp(z_j)}. \quad (2.8)$$

Softmax normalisation does not guarantee calibration. Neural networks can be overconfident, so a displayed score must not be interpreted as diagnostic certainty (Guo et al. 2017). Calibration should be evaluated on an independent validation design if the model is developed further.

Grad-CAM uses gradients flowing into a convolutional layer to form a coarse localisation map (Selvaraju et al. 2017). Such maps can support debugging and hypothesis generation, but they neither prove causal reasoning nor validate clinical relevance. This project labels them as research visualisations.

Breast ultrasound research by Yap and colleagues also shows how the task has moved from handcrafted lesion localisation and conventional classification towards end-to-end and attention-based learning. Early work addressed initial lesion detection and compared classification pipelines (Yap, Edirisinghe, and Bez 2008; Yap, Edirisinghe, and Bez 2009). Later studies evaluated convolutional detection, end-to-end lesion recognition, and region-of-interest localisation (Yap et al. 2018; Yap et al. 2019; Yap et al. 2020). AAU-Net extended this line of work to adaptive attention for lesion segmentation (Chen et al. 2023). These papers cover different tasks, so their reported scores are not interchangeable with the binary image classification result in this dissertation. Their main contribution here is methodological: localisation, preprocessing, split design, and evaluation must be specified together.

2.7 Review of public breast ultrasound datasets

Public breast ultrasound datasets differ in acquisition, annotation, reference standard, lesion mix, and file organisation. These differences affect what a model can learn and how its results should be interpreted. BUS-Set brought OASBUD, RODTOOK, UDIAT, and BUSI into a common segmentation benchmark and showed the practical value of published folds and coordinated evaluation (Thomas et al. 2023). Bristow and Yap later reviewed those sources alongside BrEaST

while studying the effect of resizing on classification (Bristow and Yap 2026). Table 2.1 summarises the five sources considered in that review. The counts are for the curated single-lesion benign and malignant subset in their manuscript; the local cohort used in this dissertation differs.

Table 2.1: Public breast ultrasound datasets reviewed by Bristow and Yap. Counts refer to their curated classification subset (Bristow and Yap 2026).

Dataset	Country	Curated images	Evidence and curation considerations
OASBUD	Poland	199	Raw radio-frequency acquisitions require B-mode reconstruction. Lesion labels include biopsy and follow-up evidence, and related views require subject grouping (Piotrkowska-Wroblewska et al. 2017).
RODLOOK	Thailand	165	The source study contains 180 images and radiologist-drawn references for malignant tumours, fibroadenomas, and cysts. Scanner overlays and duplicate handling require curation (Rodtook et al. 2018; Bristow and Yap 2026).
UDIAT	Spain	159	The public set contains a small benign and malignant cohort with radiologist contours and one scanner source. It has been used for detection and segmentation research (Yap et al. 2018; Thomas et al. 2023).
BUSI	Egypt	332	The original release contains 780 normal, benign, and malignant images with masks (Al-Dhabyani et al. 2020). Published correspondence identifies duplicates, conflicting labels, and non-breast or overlay cases that require explicit handling (Pawłowska, Karwat, and Żolek 2023).
BrEaST	Poland	252	The source contains 256 patient scans, manual lesion annotations, BI-RADS descriptors, and outcomes confirmed by biopsy or follow-up. Images came from several radiologists and scanners (Pawłowska et al. 2024).

2.7.1 Ground truth, image quality, and representation

OASBUD and BrEaST provide the clearest route from image to patient-level evidence among the datasets available to this project. OASBUD includes raw signal data, reconstructed images, lesion masks, and subject identifiers. Its reconstructed B-mode images are visibly lower in resolution and noisier than many conventional exported scans. BrEaST has manual annotations and detailed patient, image, and tumour labels; the source article reports 154 benign lesions, 98 malignancies, and four normal cases from 256 patients (Pawłowska et al. 2024). The two datasets therefore differ enough in appearance to expose some acquisition sensitivity while retaining identifiers needed for group-aware splitting.

RODLOOK and UDIAT remain useful public resources, especially for segmentation and detection comparison (Rodtook et al. 2018; Yap et al. 2018; Thomas et al. 2023). They were absent from the supplied project data. Adding them would have required a new acquisition, licensing, curation, and audit cycle after the experimental protocol had been defined. They remain suitable candidates for a later external evaluation.

BUSI was available locally, but it was not suitable for the validated experiment in its present form. Its files had been renamed to generated sample identifiers and reduced to square derivatives before this revision. The original release contains masks and three diagnostic categories (Al-Dhabyani et al. 2020), while later correspondence identifies duplicate images, label conflicts, non-breast content, and overlays that require case-level curation (Pawłowska, Karwat, and Żółek 2023). Renaming broke the link needed to repeat those published checks. It also removed the evidence needed to keep every image from one patient in a single partition. A new random split of the derivative files could therefore place related or duplicated observations on both sides of the train-test boundary.

The decision to exclude BUSI is based on study validity rather than convenience. Its additional images would increase the apparent sample size, but they would also introduce a source whose identity, curation decisions, and patient independence cannot be reconstructed from the files now held in the project. Mixing that source into the test set would make the denominator larger without making the estimate more trustworthy. BUSI remains a valuable public dataset when obtained from the original release and curated with a versioned manifest. This dissertation makes no claim about its intrinsic clinical quality; it excludes only the local derivative copy.

Image-level variation is visible before modelling. Figure 2.1 compares representative local samples and their recorded pixel dimensions. OASBUD reconstruction produces a square, grainier appearance, whereas BrEaST contains sharper exported scans with variable aspect ratios. The local BUSI files have already been reduced to square images and show repeated banding, so they cannot be treated as untouched examples of the published BUSI release. The plate records these visible differences without assigning a quality score.

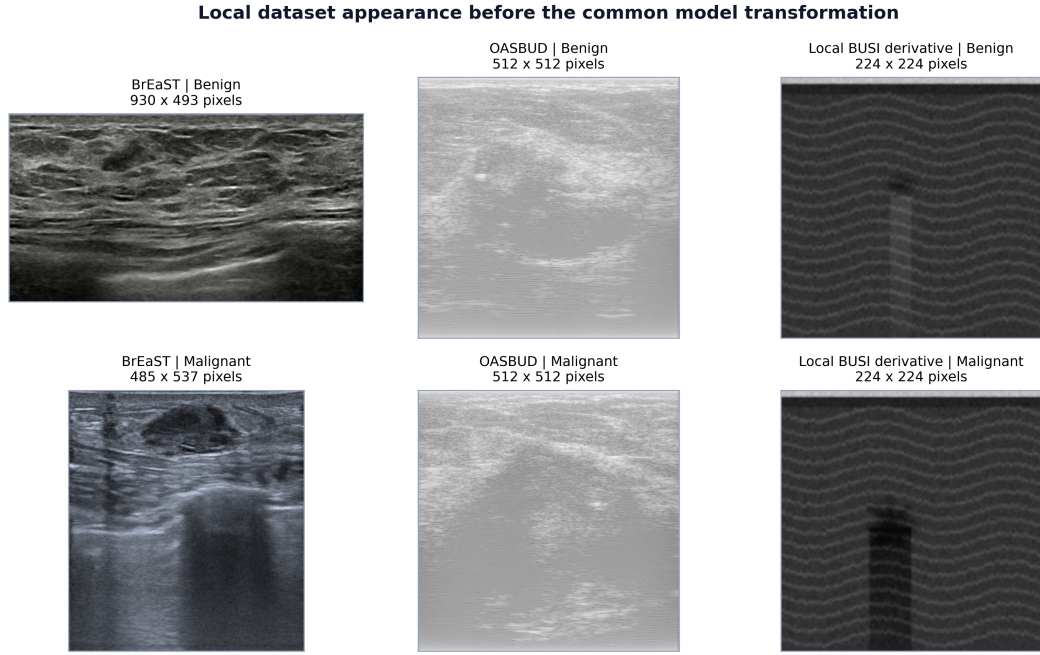


Figure 2.1: Representative benign and malignant images from the local BrEaST, OASBUD, and BUSI-derived folders. Each image retains its display aspect ratio. The BUSI panels show the processed local derivative rather than the original public release, which is why those files are excluded from validated evaluation.

2.7.2 Justification for the selected research cohort

The validated experiment uses OASBUD and BrEaST for four practical reasons. First, both retain subject identifiers that support deterministic grouping and a direct cross-partition audit. Second, both provide lesion annotations that the common region-of-interest pipeline can discover. Third, their labels have documented biopsy or follow-up support, although the proportion and form of verification differ across benign cases (Piotrzkowska-Wroblewska et al. 2017; Pawłowska

et al. 2024; Bristow and Yap 2026). Fourth, their contrasting image appearance provides a more informative local test than training and evaluating on one source alone.

This choice also has limitations. Both datasets originate in Poland, and the local audited test set contains only 62 images. The study therefore does not measure geographic generalisation, performance across the full range of scanners, or clinical prevalence. RODTOOK, UDIAT, or an independently obtained original BUSI release with verified provenance could support a later external study. That later protocol should freeze patient-level manifests before model selection and report results separately for each source rather than presenting a pooled score alone.

The review supports the evidence boundary adopted in this dissertation. Published breast ultrasound studies remain heterogeneous in population, reference standard, threshold, and validation design. A 2024 systematic review found insufficient evidence that deep learning systems outperform or consistently assist human readers in clinical breast ultrasound and called for more standardised, multicentre evaluation (Dan et al. 2024). Earlier observer research also showed that agreement varies by sonographic feature and case difficulty (Shimamoto et al. 1998). These findings support cautious reporting of the present local-cohort result.

2.8 Literature synthesis and research gap

The literature establishes several relevant components: public breast-ultrasound datasets and coordinated benchmarks (Al-Dhabyani et al. 2020; Piotrkowska-Wroblewska et al. 2017; Thomas et al. 2023; Pawłowska et al. 2024); lesion localisation and classification from conventional methods through end-to-end convolutional systems (Yap, Edirisinghe, and Bez 2008; Yap, Edirisinghe, and Bez 2009; Yap et al. 2018; Yap et al. 2019; Yap et al. 2020; Cheng et al. 2016; Zeimarani et al. 2020); efficient and residual transfer-learning architectures (He et al. 2016; Tan and Le 2019); image enhancement and filtering (Zuiderveld 1994; Yap, Edirisinghe, and Bez 2010; Osman et al. 2018); attention-based segmentation (Chen et al. 2023); and perturbation-based robustness testing (Jiang et al. 2023). Sonographic appearance can also vary with lesion size and subtype, which makes a narrowly sampled image collection a weak basis for broad clinical

claims (Yoon et al. 2018). The gap addressed here is practical integration with an explicit evidence boundary. A classifier, dashboard, and polished report can appear complete while still lacking the provenance needed to justify its headline numbers.

Accordingly, this dissertation treats data auditability as part of the modelling method. It implements the full software path, reports the current executable output as provisional, and defines the group-aware experiment required before comparative or clinical conclusions can be drawn.

2.9 Image formation and within-subject variation

In practice, the displayed image contains several transformations between the acoustic interaction and the pixel values saved to disk. Beamforming combines received signals; envelope detection converts oscillatory information into an intensity representation; log compression maps a large dynamic range into a displayable range; and post-processing may smooth, sharpen, or suppress parts of the signal. The exact sequence is device-dependent and is rarely recoverable from a public PNG file. A learning system therefore observes a measurement product rather than a neutral photograph of anatomy.

This has two consequences for the present study. First, a network may exploit acquisition cues that correlate with a label in one collection. Second, a preprocessing operation that makes an image look clearer to a person may remove a cue that the network used, or may amplify a nuisance pattern. The appropriate response is not to assume that enhancement is beneficial. It is to record the operation, compare it under controlled conditions, and inspect errors with the same transformation used at inference.

Image variability also exists within a subject. A lesion may be captured at different angles, depths, or compression states. These views are related observations, not independent samples. If one view is placed in training and another in testing, the test score estimates recognition of a familiar subject distribution more than recognition of a new patient. This is why a subject-level grouping rule is part of the experimental design rather than a post-hoc data-cleaning step.

2.10 Practical implications of preprocessing choices

The choice of crop is as important as the choice of resize. A tight lesion crop can increase the proportion of pixels occupied by the region of interest, but it can remove posterior shadowing, surrounding tissue, or the probe-related context that a radiologist would use. A wide crop retains context but makes the lesion smaller after resizing. When a mask is available, the margin around the bounding box is therefore a modelling decision rather than a neutral technical detail. The fixed margin used here is an inspectable rule, not an optimised clinical parameter.

Padding introduces a different prior. Reflection padding copies local boundary structure, reducing the hard edge produced by constant padding, but it can also create a repeated texture that was not acquired by the scanner. Centre cropping avoids invented borders but may remove an off-centre lesion. Direct resizing preserves every pixel but changes shape. These trade-offs can be described mathematically, yet their clinical relevance remains empirical. A good report should show representative transformations and retain the original-to-processed mapping so that an apparent improvement can be traced back to a concrete change in the input.

CLAHE also changes the distribution seen by a pretrained backbone. ImageNet weights were learned from natural images with a particular range of colour and local contrast. Applying a luminance transformation may make lesion boundaries easier to inspect, but it can move the input away from the statistics assumed by the initial filters. This is one reason to keep the operation configurable rather than baking it into a hidden data-export step. It allows checkpoint metadata to state whether the model saw raw or enhanced input.

2.11 Transfer learning under small-sample uncertainty

Transfer learning is not a guarantee of suitability. The source and target domains differ in colour, texture, scale, and semantic structure. Early convolutional filters may still provide useful edge and frequency responses, whereas later layers can encode object categories that are irrelevant to ultrasound. Fine-tuning all layers on a small cohort may overfit; freezing too many layers may leave a domain mismatch unresolved. A defensible experiment should state which layers are trainable, whether pretrained weights were downloaded at run time, and whether

the same schedule was applied to every comparator.

The small-data setting creates a second issue: the variance of a metric can be as important as its point estimate. With only a few dozen test images, one changed prediction moves accuracy by more than one percentage point. A ranking of three architectures on one split can therefore be unstable. The comparison is useful as a baseline and as a source of hypotheses, while a larger frozen cohort is needed for a firmer ranking.

2.12 Interpreting confidence and explanation

The interface uses an abstention threshold as a presentation choice. When the maximum class probability is below the configured threshold, the API returns an “UNCERTAIN” display prediction while preserving the underlying model prediction and probabilities. Abstention does not calibrate the model or make it safe. It prevents low-margin outputs from being presented as decisive labels. A future study should select the threshold on validation data, report coverage alongside selective risk, and test whether abstention removes ambiguous cases across datasets rather than only within one split.

Explanations should be read in the same cautious way. Grad-CAM highlights the sensitivity of a selected feature map under a particular forward and backward pass. It does not identify a lesion boundary, establish that the highlighted pixels caused the decision, or replace a reader study. Useful checks include overlaying maps on images from both classes, measuring stability under small transformations, and comparing maps for correct and incorrect predictions. These checks can reveal shortcut learning, but a visually plausible overlay remains a debugging aid rather than a clinical explanation.

2.13 Provenance as part of the literature gap

Provenance includes more than a citation to a dataset paper. A repository should record the precise archive or release used, the local transformation applied to filenames, the licence terms, and any excluded files. If a file is renamed, the original identifier should be retained in a manifest rather than discarded. If masks are copied beside images, their relationship should be checked programmatically.

If a label is collapsed, the mapping should be written down before training. These details are easy to omit because they do not appear in a model summary, yet they determine whether another researcher can reconstruct the cohort.

Evaluation design should distinguish development decisions from final measurement. Hyperparameters, preprocessing choices, and checkpoints are selected using development data. The test set is then used once to estimate performance under the predeclared protocol. Repeatedly inspecting test predictions and adjusting the model creates a hidden validation loop. On a small dataset this leakage can be subtle: a researcher may remember a difficult image, change a crop rule, and rerun the test without recording the decision. A frozen manifest and generated artefacts provide a simple defence against that drift.

2.14 From clinical descriptors to pixels

Clinical ultrasound descriptors are not pixel labels. Shape, orientation, margin, echogenicity, and posterior features are interpreted together and in relation to patient history. A classifier that receives only a cropped image has no explicit access to that history and may not observe every descriptor reliably. Pixel learning can still answer a narrower question: whether the image representation contains information correlated with the supplied class label.

The distinction also affects explanation. A saliency map that highlights a margin-like region may be consistent with a clinical description, but consistency is not validation. A map that highlights a ruler, annotation, or dark border may reveal a shortcut. Human review should therefore compare explanations with acquisition context and, where possible, with independent lesion annotations. The correct conclusion may be that a model is using an artefact, even if its numerical score is high.

2.15 Class imbalance and decision thresholds

Binary classification is often presented as if the two errors have equal consequences. In practice, the preferred operating point depends on the purpose of the system. A screening aid may prioritise sensitivity, whereas a triage tool may need a different balance of false positives and false negatives. The present project

reports threshold-free ranking through ROC-AUC and thresholded predictions at the evaluator’s default rule, but it does not select a clinical operating point.

The current report includes a precision-recall curve from the stored test probabilities. This view becomes especially important if the malignant class is less frequent in an external cohort. It should always be read with the class count and uncertainty because a curve can look persuasive while being supported by few positive cases. This is another reason to retain the per-image prediction table.

2.16 Reproducible reporting in medical imaging

Good reporting involves more than adding decimal places. It requires naming the unit of split, describing preprocessing, identifying the source of labels, and making exclusions visible. A reader should be able to answer which images were used, which were withheld, and whether any subject appeared in more than one partition. The audit and manifest in this project address those questions.

The literature review therefore supports a restrained conclusion. CNNs, transfer learning, CLAHE, and saliency methods are all reasonable tools for exploration, but none removes the need for provenance. The most transferable lesson is procedural: a result becomes more credible when a second person can reproduce it from the stored configuration and see where uncertainty enters the chain.

Chapter 3

Methodology and Experimental Design

3.1 Study design and evidence levels

The project follows an implementation-and-evaluation design. Its primary unit of analysis is an image, but its required unit of partitioning is the subject. Two evidence levels are distinguished:

Engineering evidence confirms that preprocessing, model loading, inference, metric computation, and artifact generation execute consistently.

Generalisation evidence estimates performance on subjects independent of model development and requires verified subject-level separation, preserved provenance, and a frozen test set.

The current repository supports engineering evidence and a local patient-separated evaluation for the repaired BrEaST and OASBUD cohort. BUSI is excluded because the local derivative lacks traceable identifiers. The result remains local-cohort research evidence rather than external or clinical validation.

3.2 Data sources and label mapping

The local project combines organised copies of BUSI, OASBUD, and BrEaST. BUSI is a public B-mode image dataset with normal, benign, and malignant labels and lesion masks (Al-Dhabyani et al. 2020). OASBUD contains multiple views per lesion (Piotrzkowska-Wroblewska et al. 2017). The BrEaST copy uses case identifiers and includes a small normal subset.

The configured task is binary: benign and malignant. Files in a normal directory are mapped to the non-malignant class by the loader. This makes the executable task consistent but collapses clinically different categories, so the decision must be reviewed and predeclared before a definitive experiment.

The validated loader finds the counts in Table 3.1. Mask files are excluded from the image count; BUSI is retained for provenance context but excluded from validated training.

Table 3.1: Images in the repaired validated local folders. BUSI is excluded because its patient identifiers are unavailable.

Dataset	Training	Validation	Test	Total
BUSI	n/a	n/a	n/a	n/a
OASBUD	138	30	30	198
BrEaST	164	33	32	229
Validated combined	302	63	62	427

3.3 Split-integrity audit

The read-only audit enumerates eligible image files, derives subject identifiers where the naming convention permits, and records every partition associated with an identifier. For OASBUD it removes the view suffix; for BrEaST it uses the caseNNN prefix. The BUSI files in this local copy were renamed to generated sample identifiers, so patient separation cannot be reconstructed from filenames alone.

An audit passes only when every included dataset has verifiable identifiers and no identifier occurs in more than one partition. The repaired OASBUD and BrEaST folders pass this check. BUSI remains unverifiable and is excluded from the validated run. The training entry point stops by default if a configured dataset fails; an `--allow-unaudited-data` override exists solely for explicitly provisional engineering runs and is recorded in the checkpoint through the audit status.

The `rebuild_subject_splits.py` utility copies grouped source files into deterministic subject partitions and writes a manifest. Copying rather than moving preserves the legacy source dataset. The generated manifest and audit report should be retained with run metadata, subject to dataset licensing constraints.

3.4 Preprocessing pipeline

All training, evaluation, command-line prediction, and API routes call the same preprocessing implementation. Images are decoded to RGB, optionally enhanced with CLAHE, transformed according to a selected geometric strategy, converted to tensors, and normalised with ImageNet channel statistics for the transfer-learning backbones.

Figure 3.1 summarises the implementation path. The diagram is a software view of the experiment: the audit determines which files can enter the validated cohort, preprocessing defines the tensor presented to the model, and the evaluator writes the evidence consumed by the report and dashboard. Keeping these stages explicit reduces the risk that a browser demonstration silently uses a different transformation from the reported checkpoint.

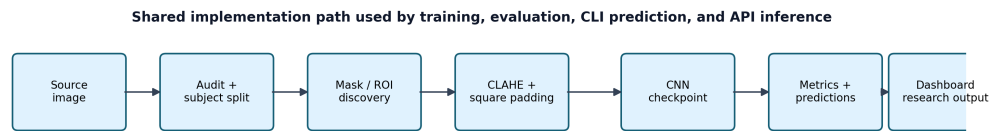


Figure 3.1: Implementation path shared by training, evaluation, command-line prediction, and API inference.

3.4.1 CLAHE

The image is converted from RGB to LAB. CLAHE is applied only to the luminance channel using OpenCV with clip limit 2.0 and a 16×16 tile grid; the result is then converted back to RGB. The operation can be disabled through configuration or request parameters.

3.4.2 Mask discovery and region selection

The loader recognises paired masks ending in `_mask.png`, `_tumor.png`, or `_lesion_mask.png`. When a non-empty mask is available, the inclusive lesion bounding box is expanded by a 20% width and height margin, clipped to the image boundaries, and extracted. When no mask is available, the implementation

uses the central 80% of the image before margin expansion. This fallback is deterministic but is not lesion localisation and is documented as such.

3.4.3 Square padding

For a crop of height h_c and width w_c , let

$$d = \max(h_c, w_c). \quad (3.1)$$

The vertical and horizontal padding are

$$p_t = \left\lfloor \frac{d - h_c}{2} \right\rfloor, \quad p_b = d - h_c - p_t, \quad (3.2)$$

$$p_l = \left\lfloor \frac{d - w_c}{2} \right\rfloor, \quad p_r = d - w_c - p_l. \quad (3.3)$$

OpenCV `BORDER_REFLECT_101` fills the boundary, after which the square is resized to 224×224 pixels with bicubic interpolation. Because both output axes scale from side d , the final resize is isotropic. Direct resize and centre crop are implemented as comparator strategies.

3.4.4 Augmentation

Training uses random horizontal flips and rotations up to 15° . Vertical flips are intentionally excluded because the depth axis has acquisition meaning in ultrasound. Validation and test transforms contain no random augmentation.

3.5 Model architectures

Three architectures share a two-class output contract:

- EfficientNet-B0 initialised from torchvision ImageNet weights, with its classifier replaced by dropout and a two-output linear layer (Tan and Le 2019);
- ResNet-50 initialised from torchvision ImageNet weights, with its fully connected layer replaced by dropout and a two-output linear layer (He et al. 2016); and

- Custom CNN with four convolutional stages, batch normalisation, ReLU, max pooling, adaptive average pooling, dropout, and a two-output linear layer.

Transfer learning is used only during training. Evaluation reconstructs the architecture without downloading pretrained weights and loads the stored state dictionary.

Figure 3.2 shows the implementation-level comparison. The blocks are schematic rather than parameter-count diagrams; their purpose is to show the different representation pathways that lead to the same two-class output. EfficientNet-B0 and ResNet-50 use ImageNet initialisation, while the custom CNN learns its convolutional representation from the project data. All three are evaluated through the same loader and metric writer.

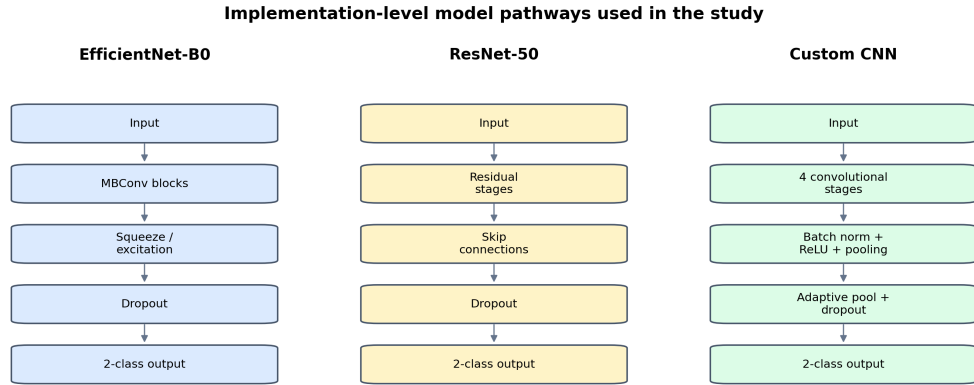


Figure 3.2: Schematic implementation pathways for the three model families.

3.6 Optimisation and reproducibility controls

The default optimisation configuration uses class-weighted cross-entropy, AdamW with learning rate 10^{-4} and weight decay 10^{-4} , cosine annealing, batch size 16, up to 25 epochs, and early stopping after five validation epochs without loss improvement. Class weights are computed from the training directory counts as

$$w_c = \frac{N}{K N_c}, \quad (3.4)$$

where N is the training size, K is the number of classes, and N_c is the count for class c .

The training command accepts a seed, defaulting to 42, and seeds Python, NumPy, PyTorch, and CUDA where available. Each best-loss checkpoint stores the architecture, epoch, optimiser state, validation loss and accuracy, class names, seed, learning rate, batch size, requested epoch budget, dataset-combination flag, crop strategy, CLAHE flag, creation time, and data-audit status.

Exact bitwise reproducibility can still depend on hardware, library versions, and non-deterministic kernels. A final run should therefore record the environment lock, device, repository revision, and checksums in addition to the checkpoint metadata.

3.7 Evaluation protocol

The evaluator loads a named checkpoint and processes validation or test data without random augmentation. For each sample it records the relative path, true class, predicted class, and malignant-class probability. It then writes:

- predictions.csv, the per-image evidence table;
- metrics.json, including accuracy, macro precision, macro recall, macro F1, ROC-AUC, confusion matrix, preprocessing configuration, generation time, audit status, and audit checksum;
- confusion_matrix.png; and
- roc_curve.png.

Accuracy is

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3.5)$$

and class F1 is

$$F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3.6)$$

Macro metrics average class-level values equally. ROC-AUC is calculated from malignant-class probabilities for this binary task. The current post-hoc analysis adds confidence intervals and a descriptive reliability diagram from the stored test

probabilities. A definitive experiment would predeclare exclusions and missing-data handling, fit any calibration method on validation data, and preserve the test set for one final evaluation.

3.8 Web research demonstration

FastAPI serves model inference and a static interface. Upload validation limits files to recognised image formats and 20 MB. The client exposes the selected pre-processing configuration, probability vector, Grad-CAM overlay, image-quality heuristics, batch outputs, and controlled seeded perturbations. It also reads the evaluator’s metrics file rather than embedding benchmark values.

The interface does not assign a diagnosis, BI-RADS category, clinical priority, or management recommendation. Its report text and batch labels explicitly identify outputs as research-only. Ground truth is accepted only when supplied by the dataset endpoint or the user; it is never inferred from a filename.

Figure 3.3 shows the implemented single-image workspace after loading a labelled example. The persistent navigation separates prediction, noise, batch, evidence, dataset, and documentation tasks. The header keeps the selected model and preprocessing controls visible, while the main panel places the source image beside the Grad-CAM inspection layer. A yellow evidence banner tells the viewer that the results are provisional. This layout makes the experimental configuration and evidence status harder to overlook during a demonstration.

Figure 3.4 records the complete browser-facing workflow. The single-image page supports case inspection; the noise page applies a seeded perturbation to one image; the batch page processes labelled research examples; the evidence page reads generated metrics and the split audit; the dataset page exposes folder contents with a provenance warning; and the documentation page states appropriate and unsupported use before presenting the project record. These pages share one model and preprocessing configuration bar so a demonstrator can see the active settings.

3.9 Definitive experiment specification

Before comparative claims are reported, the following sequence is required:

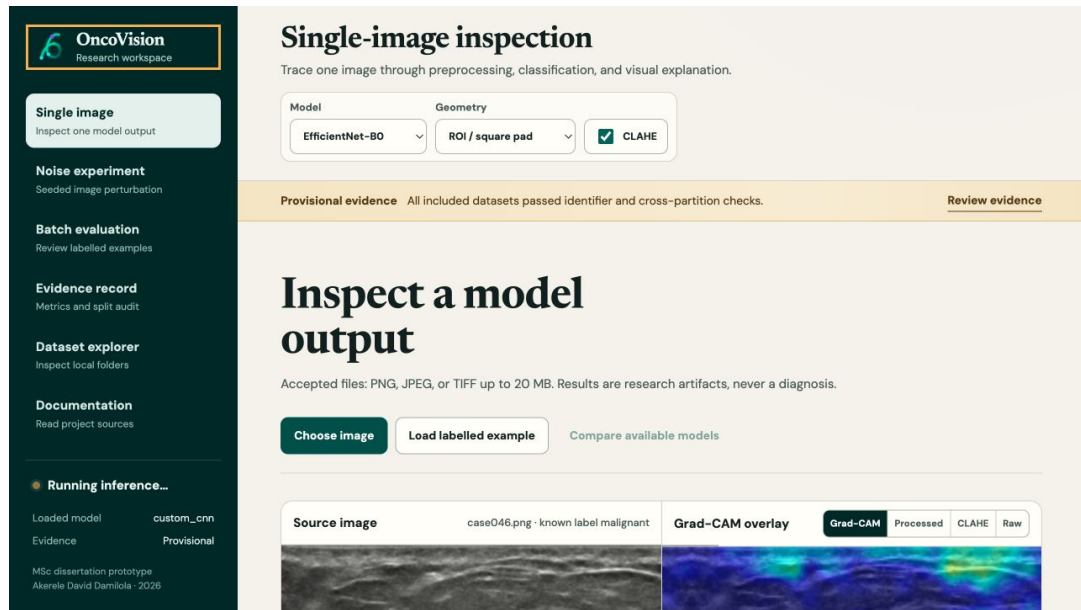


Figure 3.3: Implemented OncoVision research dashboard showing the single-image inspection workspace, model and preprocessing controls, source image, and Grad-CAM layer. The interface identifies the evidence as provisional and the output as a research artefact.

1. recover original identifiers and document licences, versions, exclusions, and label mapping;
2. assign all images, views, and masks from one subject to exactly one partition;
3. freeze the test manifest before model selection;
4. select preprocessing and hyperparameters using training and validation data only;
5. train each comparator across predeclared seeds and equal tuning budgets;
6. evaluate locked checkpoints once on the untouched test subjects;
7. publish per-image predictions, aggregate metrics with uncertainty, audit outputs, and failure analysis; and
8. regenerate all dissertation tables and figures directly from those artifacts.

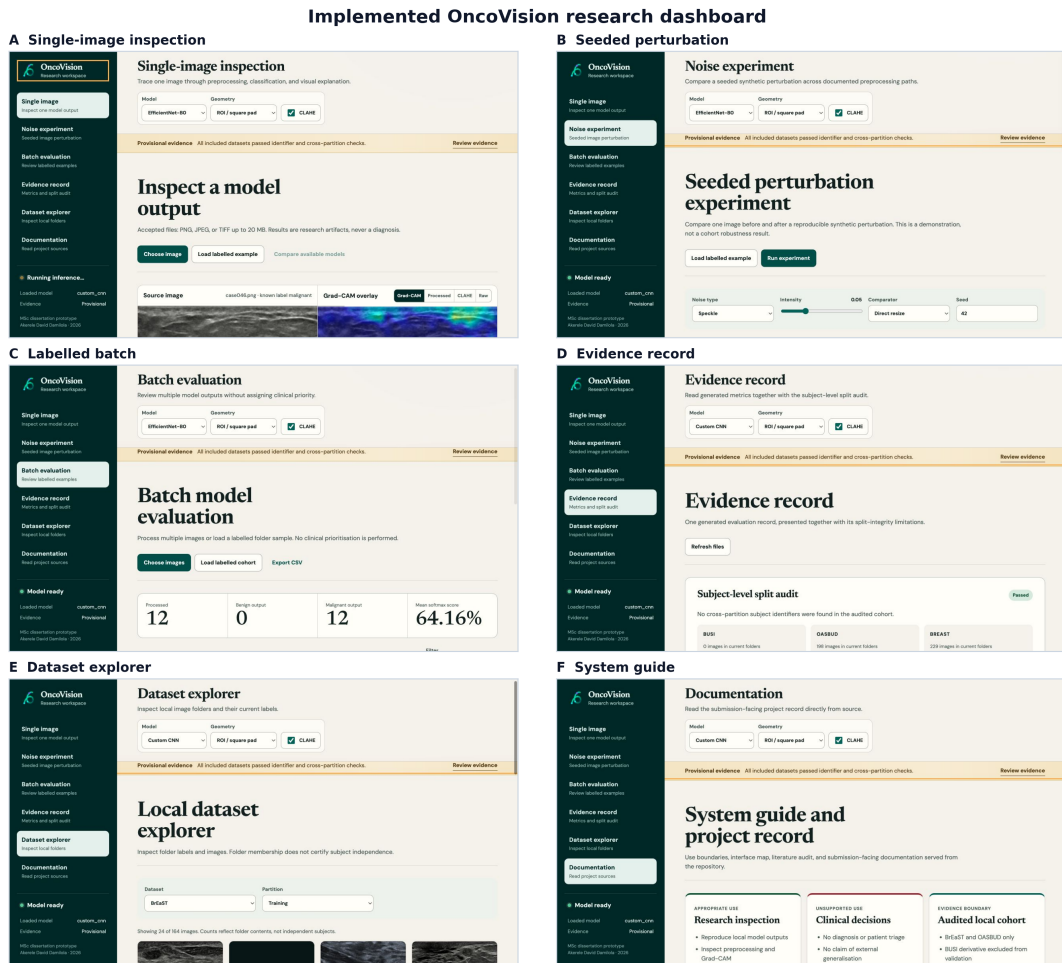


Figure 3.4: Browser-verified tour of the implemented dashboard. Panels A to F show single-image inspection, a seeded perturbation demonstration, labelled batch output, generated evidence, the local dataset explorer, and the research-use guide. The screens are documentation of the software, not additional model-performance evidence.

This sequence converts the implemented framework into a valid generalisation study while preserving the distinction between exploratory development and final evidence.

3.10 Operational data curation

The audit is intentionally read-only during ordinary evaluation. It enumerates eligible files, applies dataset-specific identifier parsers, and records every partition associated with each identifier. A report that only counts files can appear balanced while hiding duplicate subjects; a report that only checks folder names can miss a duplicated identifier under a different filename. The audit output is therefore a table of observations, not merely a pass/fail flag.

For the current cohort, the BrEaST parser retains the case prefix and the OASBUD parser removes the view suffix. These identifiers are used for grouping and audit purposes, not as model features. The manifest should be treated as sensitive research metadata. It may be committed in an anonymised form when the dataset licence permits, while a private mapping to source records is retained by the research team. This supports reproducibility without unnecessarily redistributing patient-linked information.

Label mapping deserves the same attention as grouping. The binary task treats benign and malignant as the output classes and maps a normal directory to the non-malignant class. That convention makes the loader compatible with heterogeneous folders, but it combines a normal screening appearance with a benign lesion. A definitive study could use three classes, exclude normals, or define a clinically motivated two-class question. The choice should be made before inspecting test results because it changes the meaning of precision, recall, and error severity.

3.11 Preprocessing as a versioned transform

The shared preprocessing function is a contract between the dataset and every consumer of a checkpoint. It should be possible to take one source image, apply the stored configuration, and obtain the same tensor during training, evaluation,

command-line prediction, and web inference. Configuration values such as image size, CLAHE state, crop strategy, padding mode, normalisation constants, and augmentation policy are therefore recorded rather than inferred from the interface.

The transform is deterministic after random training augmentation has been separated from validation and test. This makes failure analysis practical: an investigator can reproduce the pixels that generated a probability, compare them with the original image, and test a proposed change without adding a new random factor. It also makes the dashboard honest. A selected direct-resize option must change the operation used by the evaluator, not merely change a label in the browser.

The fallback crop is particularly important to document. When a mask is absent, the implementation uses a central region as a conservative placeholder. The lesion may be elsewhere, so the crop is not a lesion localisation method. It keeps the research pipeline executable while making missing annotation visible. A production-oriented implementation would require a validated detector, an explicit no-mask policy, or human review instead of silently treating the centre as ground truth.

3.12 Model comparison and training controls

The three architectures represent different capacity and inductive-bias choices. EfficientNet-B0 offers a compact transfer-learning backbone; ResNet-50 provides a deeper residual comparator; and the custom CNN supplies a small from-scratch baseline. A comparison is meaningful only when the surrounding conditions are aligned. The same subject partitions, input size, class definition, early-stopping rule, and evaluation script should be used for every model. If one network receives more tuning or a different augmentation policy, the result is a comparison of complete recipes rather than architectures.

Class-weighted cross-entropy gives a rare class more influence, but it does not create new clinical information and can increase variance when counts are small. Weights must therefore be computed from the training partition only. Validation and test labels must not affect the loss, threshold, or checkpoint selection. This separation is recorded in training metadata and should be checked whenever a run is reproduced.

The seed is a reproducibility control, not a guarantee of certainty. It fixes intended pseudo-random streams, yet hardware, library versions, and parallel kernels may still change final weights. The recommended record includes Python and PyTorch versions, device type, repository revision, source-data checksum, and checkpoint hash. These details make it possible to explain a small rerun difference rather than treating it as an unexplained modelling improvement.

3.13 Metric interpretation

Accuracy is intuitive but can conceal class imbalance. Macro recall gives each class equal weight and is useful when errors in either class matter. ROC-AUC measures ranking across thresholds, whereas a deployed decision requires one operating threshold. Calibration measures whether predicted probabilities correspond to observed frequencies; a model can have reasonable AUC and still be overconfident. The evaluator therefore stores probabilities and per-image predictions so that additional metrics can be calculated without retraining.

Uncertainty intervals are particularly important for a small test partition. A bootstrap interval is not a substitute for a larger sample or external site, but it communicates how much a point estimate would move under resampling. It should be described as an uncertainty summary for sampled images, not a guarantee about future patients. A final protocol should predeclare the interval method, number of resamples, and treatment of degenerate class samples.

3.14 Web interface and software verification

The web application is a research instrument for making the pipeline inspectable. Upload validation limits input type and size, while the API exposes preprocessing choice and separates the raw model prediction from the display-level abstention label. Batch mode reads the same model and transformation as single-image inference. The metrics view reads generated evaluator artefacts instead of embedding a manually typed score, reducing the chance that dashboard and dissertation diverge.

Software tests check contracts that would otherwise fail silently: processed

dimensions, paired-mask discovery, audit status in outputs, and the research-only API wording. These tests do not prove medical validity. They provide a minimum guard against accidental changes to the data path while scientific questions remain under study. A seeded brightness, contrast, or noise change is likewise a demonstration until it is applied to a held-out set and reported with prediction-level metrics and uncertainty.

3.15 Run management and artefact lineage

Each experiment should be treated as a small bundle rather than a single checkpoint. The bundle contains the configuration, audit report, manifest reference, model state, metrics, predictions, plots, and environment information. Naming the bundle with a run identifier prevents a later evaluation from overwriting the files used to write the dissertation. It also makes it possible to compare a provisional run with a corrected run without relying on memory.

The evaluator’s checksum is a lightweight lineage mechanism. It does not prove that the underlying public dataset has not changed, but it records which audit output was associated with a result. In a final submission, the checksum should be accompanied by a repository revision and a data-release identifier. When licences prevent redistribution of images, these metadata still allow an authorised collaborator with the same source archive to verify the run.

3.16 Reproducible training sequence

A complete training sequence begins with a clean audit, writes the selected dataset list, sets the seed, constructs the dataloaders, and records the model and optimiser configuration before the first update. During training, validation loss and accuracy are written at every epoch so that early stopping can be inspected. The selected checkpoint is the one defined by the stopping rule, not the last file written to disk. Evaluation then loads that checkpoint with random augmentation disabled.

This sequence avoids a common ambiguity in student projects, where a reported model may have been trained for one number of epochs but evaluated from another saved state. It also separates model selection from test measurement. If

a later change improves validation loss, the old test result remains intact and can be compared as a previous run rather than silently replaced.

3.17 Ethical handling of uncertainty

The interface wording is part of the methodology because users interpret labels in context. Research-only language, visible preprocessing settings, and the abstention state reduce the risk that a demonstration is mistaken for a medical recommendation. The application should also make it easy to inspect the original image and generated overlays rather than presenting a single coloured badge as if it were a diagnosis. These are modest human-factors controls, but they are appropriate for the scope of the project.

An ethical presentation also avoids implying that a low-confidence image is unimportant. Abstention means that the model’s output is not decisive under the configured rule; it does not mean that the underlying case is benign or safe to ignore. Documentation should say this explicitly and should discourage use outside the research context.

Chapter 4

Results, Reproducibility, and Evaluation Status

4.1 Purpose of this chapter

This chapter distinguishes executable evidence from legacy illustrative material. Performance values are reported only when they are generated by the repository’s evaluation script and accompanied by a split-integrity assessment. This distinction is essential because an image-level split can produce optimistic estimates when scans or views from the same subject occur in more than one partition.

4.2 Current cohort audit

The validated loader uses repaired BrEaST and OASBUD subject-level partitions, containing 302 training, 63 validation, and 62 test images. The audit reports no cross-partition subjects in either dataset. BUSI remains excluded because its generated `sample_` filenames prevent verification of patient-level separation from the local files alone.

4.3 Executable checkpoint result

The fresh EfficientNet-B0 checkpoint was evaluated with the shared preprocessing implementation on the repaired test partition. Table 4.1 is generated from that execution and describes local-cohort research evidence, not clinical or external validation.

At class level, the benign class (33 images) had precision 0.72, recall 0.64, and F1 0.68. The malignant class (29 images) had precision 0.64, recall 0.72, and F1 0.68. These values describe this held-out local cohort and do not establish

Table 4.1: EfficientNet-B0 result on the audited 62-image BrEaST and OASBUD test partition.

Model	Samples	Accuracy	Macro F1	Macro Recall	ROC-AUC
EfficientNet-B0	62	67.74%	0.6774	0.6803	0.7419

external or clinical performance.

Table 4.2: Class-level results at the fixed 0.5 malignant-probability threshold.

True class	Support	Precision	Recall	F1
Benign	33	0.7241	0.6364	0.6774
Malignant	29	0.6364	0.7241	0.6774

The confusion matrix in Figure 4.1 contains 21 correctly classified benign images, 21 correctly classified malignant images, 12 benign images classified as malignant, and 8 malignant images classified as benign. Figure 4.3 plots the corresponding threshold sweep. Both figures are generated by `evaluate.py` from the same prediction run as Table 4.1.

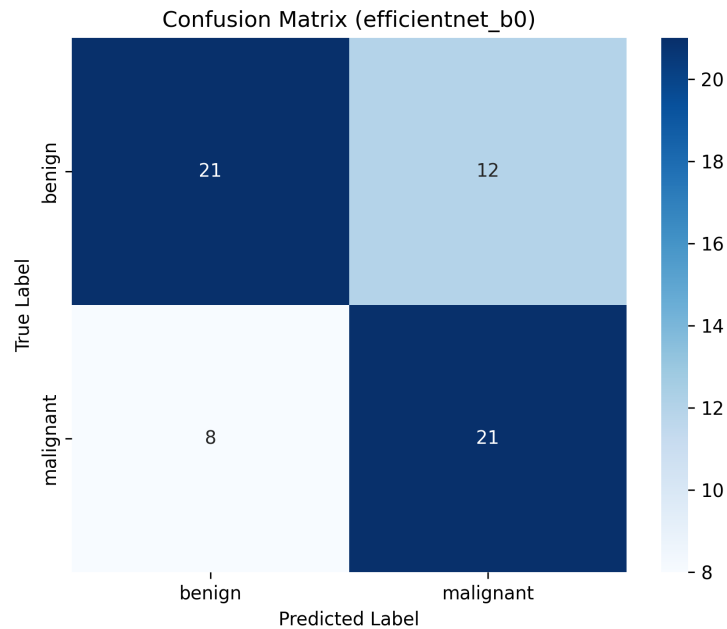


Figure 4.1: Confusion matrix for the EfficientNet-B0 checkpoint on the audited 62-image test partition. Local-cohort research evidence only.

The stored probabilities also support threshold-sensitive analysis without fitting the model again. Figure 4.2 combines a malignant-class precision-recall curve with a five-bin reliability diagram. Average precision is 0.6803, compared with a malignant prevalence of 0.4677. The Brier score is 0.2144 and the evaluator’s expected calibration error is 0.0652. These are descriptive test-set diagnostics. No probability calibration or threshold selection was fitted on the test data.

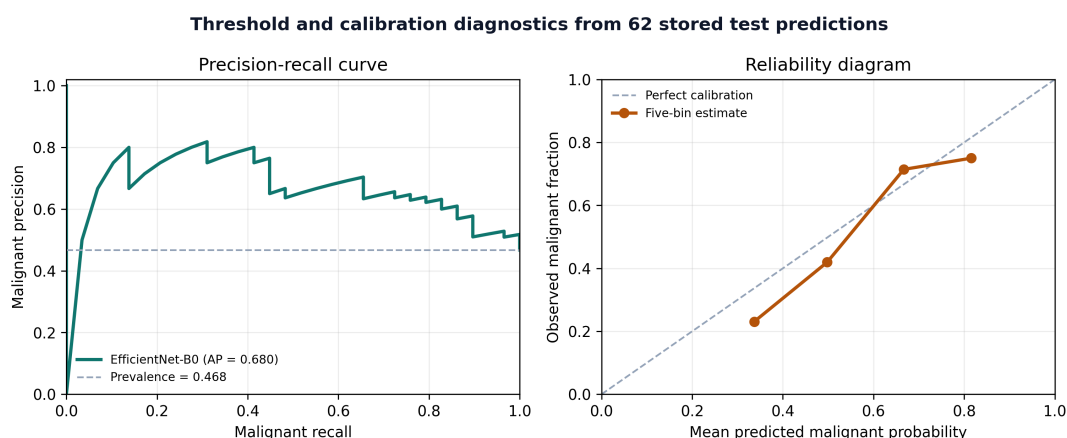


Figure 4.2: Precision-recall and reliability views derived from the 62 stored EfficientNet-B0 predictions. Average precision summarises malignant-class ranking, while the reliability panel compares mean predicted probability with the observed fraction in five bins. The small number of cases makes each point uncertain.

4.4 Withdrawn legacy figures and tables

Earlier project assets contained fixed benchmark values, simulated pseudo-labelling tables, and illustrative noise trajectories. They are not used as results because they were not regenerated from stored model predictions on an audited cohort. The superseded files have been removed from the submission-facing figures and archived with an explicit warning; they must not be cited as measured performance.

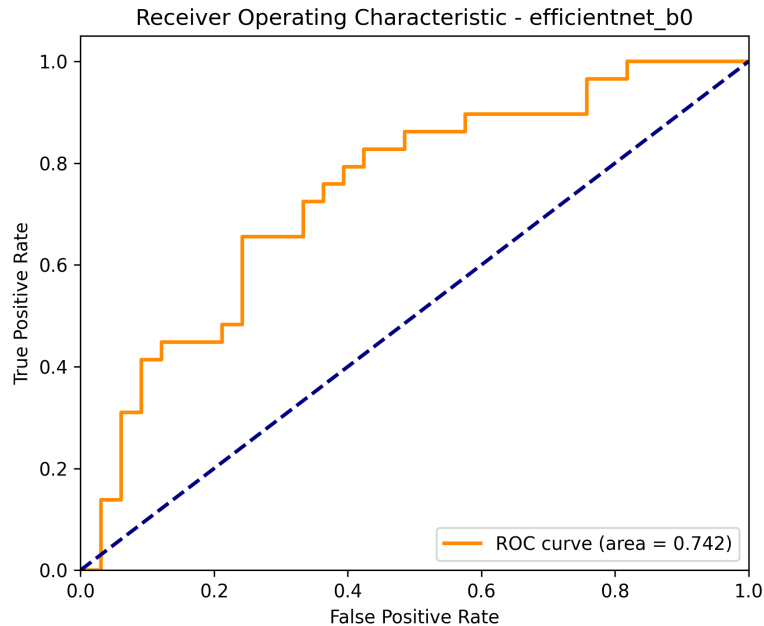


Figure 4.3: ROC curve for the same 62-image audited evaluation run (ROC-AUC 0.7419).

4.5 Additional evidence figures

The notebook contains the same confusion matrix and ROC curve included above. Three additional figures were generated from the notebook’s machine-readable evidence files rather than drawn by hand. Figure 4.4 shows the class composition of the two audited datasets across the three partitions. BrEaST contributes 229 images and OASBUD contributes 198; the test split contains 62 images in total. The figure also makes the normal category visible in the training and validation folders, even though the current binary output contract maps it to the non-malignant side of the task.

Figure 4.5 compares the three architectures under the same local test protocol. EfficientNet-B0 has the highest point estimates for accuracy, macro F1, and ROC-AUC in this run. The figure should be read as an internal benchmark, not as a comparison with published external systems. The common split and evaluator make the comparison useful for deciding what to investigate next, while the sample size limits the strength of any ranking claim.

Figure 4.6 places the accuracy values beside their bootstrap intervals. The intervals overlap substantially, so the point-estimate ordering should not be

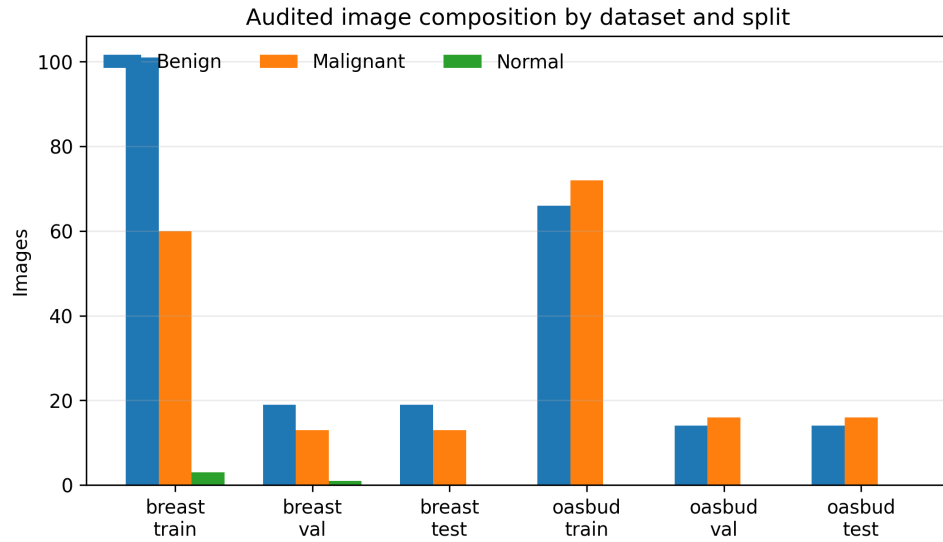


Figure 4.4: Class composition of the audited BrEaST and OASBUD folders. Counts are read from the split audit; BUSI is excluded from the validated run.

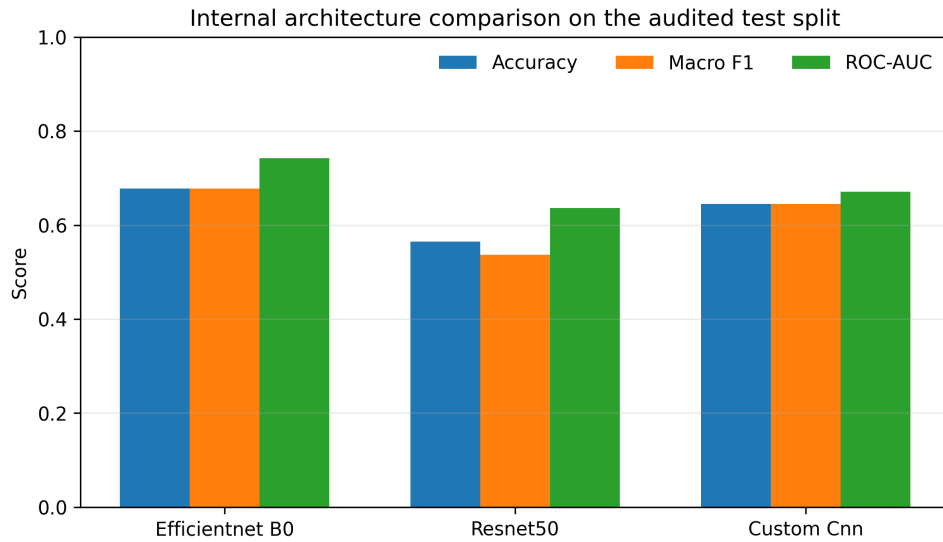


Figure 4.5: Internal same-split comparison of the implemented model families. Scores are generated from the stored benchmark JSON.

treated as evidence that EfficientNet-B0 is superior in the wider population. This is a useful scientific finding in its own right: the current data support a baseline and a protocol, while a larger subject-level cohort is needed to resolve model differences.

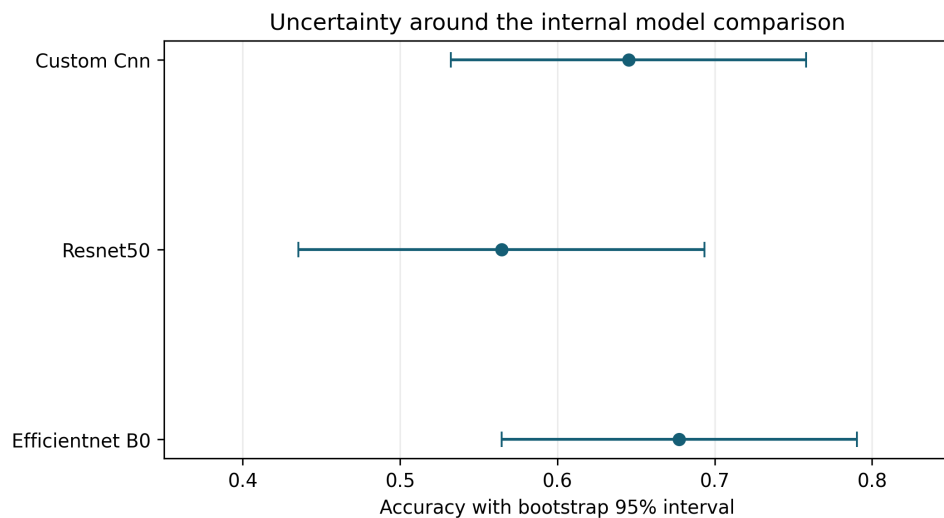


Figure 4.6: Accuracy point estimates and bootstrap 95% intervals for the internal model comparison.

Together, the figures show why the report presents the model as a research baseline. The audited class counts are small, the classes are not perfectly balanced, and the uncertainty intervals are wide. The evidence supports further experimentation with restored provenance and multi-seed evaluation rather than a claim of diagnostic readiness.

4.6 Final evaluation protocol

The final study must assign every subject, including all views and masks, to exactly one partition. It should preserve a versioned manifest with source identifiers, label, split, checksum, preprocessing configuration, random seed, checkpoint hash, and per-image prediction. Accuracy, sensitivity, specificity, ROC-AUC, precision-recall AUC, calibration, and confidence intervals should then be calculated once on an untouched test partition. Noise and preprocessing ablations require separate trained/evaluated runs rather than manually specified curves.

4.7 Interpretation

The present evidence supports the implementation of a reproducible experimental framework, not superiority of a particular architecture or preprocessing method. The results should therefore be interpreted as a baseline engineering validation. Claims of clinical robustness, optimal pseudo-labelling thresholds, or state-of-the-art diagnostic performance are deferred pending the audited experiments described above.

4.8 Internal model comparison

The same evaluator compared the three implemented architectures on the same audited test partition. EfficientNet-B0 produced the strongest point estimates in this local run, followed by the custom CNN and then ResNet-50. This ordering is descriptive only. It is based on 62 test images, uses one training seed, and does not include a multi-seed uncertainty analysis for the architecture ranking.

Table 4.3: Internal comparison on the same audited 62-image test partition. Parentheses give bootstrap 95% intervals.

Model	Accuracy	Macro F1	ROC-AUC
EfficientNet-B0	0.6774 (0.5645, 0.7903)	0.6774 (0.5589, 0.7889)	0.7419 (0.6217, 0.8616)
Custom CNN	0.6452 (0.5323, 0.7581)	0.6452 (0.5171, 0.7575)	0.6708 (0.5330, 0.7979)
ResNet-50	0.5645 (0.4355, 0.6935)	0.5374 (0.4038, 0.6649)	0.6364 (0.5054, 0.7701)

The comparison illustrates why a single accuracy value is insufficient. The EfficientNet result has a ROC-AUC of 0.7419, indicating useful ranking behaviour in this sample, but the score does not identify a clinically appropriate threshold. The custom CNN is higher than ResNet-50 on all three listed point estimates in this run. That observation is worth reporting, but it is not a scientific breakthrough: both models were tested on the same 62-image partition, each comparison represents one training seed, and every pair of bootstrap intervals overlaps. The comparison file retains only per-model summaries rather than paired predictions, so a paired test such as McNemar’s test cannot be reconstructed. A significance claim would be unsupported. The observed differences may reflect capacity, optimisation, or random variation.

4.9 Training-record limitation

The three checkpoints preserve the selected epoch and validation summary, but not the complete epoch-by-epoch history. Table 4.4 reports everything that can be recovered. A learning curve cannot be reconstructed from one selected point, so no synthetic curve is included.

Table 4.4: Checkpoint selection metadata retained by the completed runs.

Model	Selected epoch	Validation loss	Validation accuracy
EfficientNet-B0	5	0.6107	0.6667
Custom CNN	3	0.6431	0.6349
ResNet-50	2	0.6662	0.6349

The training program now writes loss, accuracy, and learning rate after every epoch for future runs and embeds the history available at each best checkpoint. This change improves the next experiment without retraining or altering the evidence reported here.

4.10 Uncertainty and calibration

For the EfficientNet checkpoint, the bootstrap 95% interval for accuracy was $[0.5645, 0.7903]$, for macro F1 $[0.5589, 0.7889]$, and for ROC-AUC $[0.6217, 0.8616]$. The expected calibration error was 0.0652 under the evaluator’s binning procedure. These values make the result more informative than a point estimate alone, while still leaving substantial uncertainty. The interval width is a direct reminder that a small test set cannot support fine-grained claims about superiority.

Calibration should be separated from abstention in the web application. The API can hide low-margin outputs behind an “UNCERTAIN” display label, but that rule does not recalibrate the underlying probabilities. The reliability diagram in Figure 4.2 is an evaluation of the saved test predictions, not a fitted recalibration model. Temperature scaling or another calibration method would need to be fitted on validation data and checked once on an untouched test set. Until then, probabilities are useful for ranking and inspection, not for estimating a patient’s probability of disease (Guo et al. 2017).

4.11 What the errors show

The EfficientNet confusion matrix contains eight malignant images classified as benign and twelve benign images classified as malignant. The two error directions are not interchangeable in a clinical setting, but the current study does not contain the clinical information needed to assign different costs or choose an operating point. It is therefore safer to report both class-wise recalls than to describe the model as simply accurate or safe.

Each mistaken image should be reviewed alongside its original file, processed crop, mask status, probability, and dataset source. Reviewers can ask whether the error is associated with an absent mask, a small lesion, a particular acquisition style, a label ambiguity, or an artefact at the image boundary. Such observations can generate hypotheses for the next experiment, but they should not be converted into causal explanations from visual inspection alone.

Figure 4.7 adds an image-level view of the stored predictions. The examples are selected by a fixed rule from `outputs/predictions.csv`: the lowest-probability correct benign case, the highest-probability correct malignant case, the highest-probability false positive, and the lowest-probability false negative. Each panel uses the same mask-guided crop and CLAHE path as evaluation. Green frames mark correct classifications and red frames mark errors.

The false positive comes from OASBUD and the false negative from BrEaST, so the two displayed errors also span the distinct visual styles seen in Figure 2.1. That pattern is useful for planning a source-stratified analysis. Four selected images cannot show whether one dataset is systematically harder, and the error causes remain unresolved without a larger sample and expert review.

The full per-image table allows a source-level check. Table 4.5 and Figure 4.8 show that performance differs markedly between the two local sources. On BrEaST, accuracy was 0.7813 and specificity 0.8947. On OASBUD, accuracy was 0.5667 and specificity 0.2857, while malignant recall was higher at 0.8125. This pattern is compatible with source-dependent acquisition or reconstruction effects, but the sample is too small to identify a cause.

OASBUD accounts for ten of the twelve false positives and three of the eight false negatives. BrEaST accounts for two false positives and five false negatives. The result suggests that a single pooled accuracy hides different error profiles. It does not show that one dataset is better, because source, case mix, reconstruction,

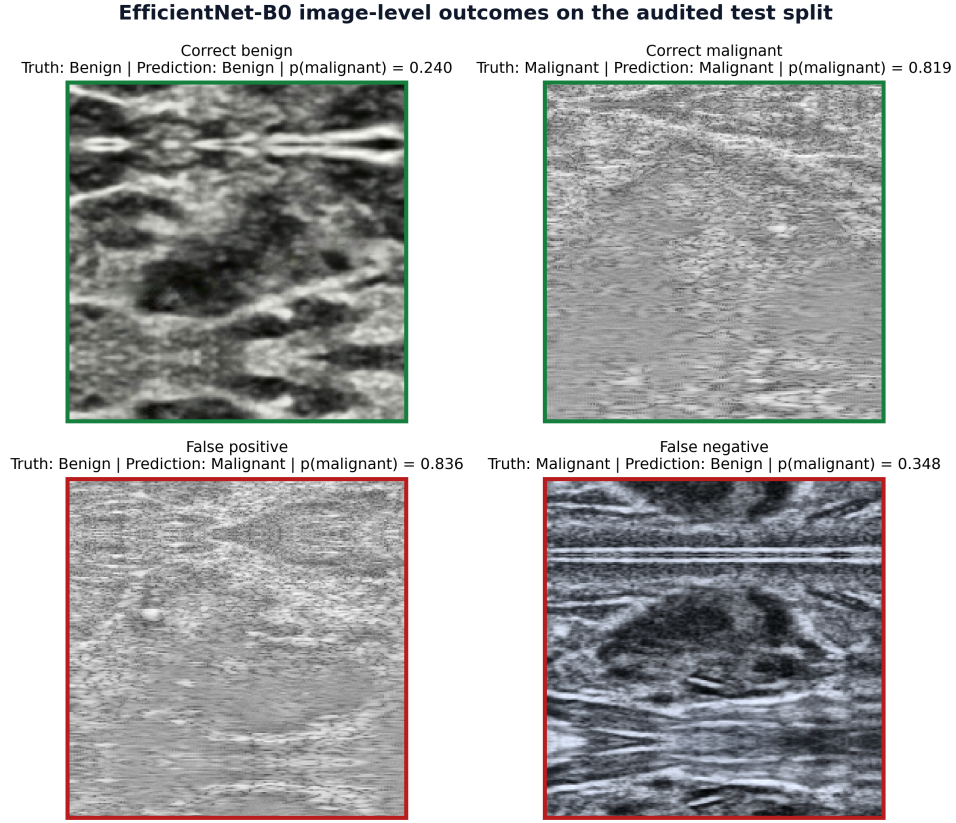


Figure 4.7: Deterministically selected correct and incorrect EfficientNet-B0 outputs from the audited test partition. Panel headings report the true label, predicted label, and stored malignant-class probability. Visual inspection supports error analysis but does not establish the clinical reason for a decision.

Table 4.5: Source-stratified EfficientNet-B0 test results at the fixed 0.5 threshold.

Source	n	Accuracy	Malignant recall	Specificity	ROC-AUC	AP
BrEaST	32	0.7813	0.6154	0.8947	0.8259	0.7840
OASBUD	30	0.5667	0.8125	0.2857	0.6786	0.6853

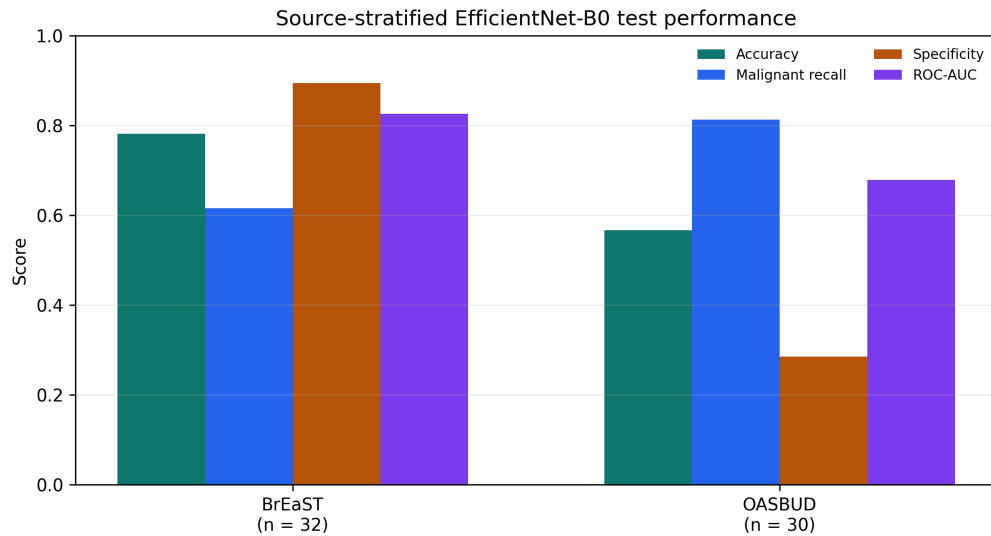


Figure 4.8: Performance separated by source for the stored EfficientNet-B0 test predictions. This is an internal diagnostic of two small subsets, not external validation or a ranking of the source datasets.

lesion characteristics, and sample size are confounded. A larger protocol should report each source separately and predefine whether thresholds are shared or source-specific.

4.12 Evidence status

The most defensible result of this chapter is a clear evidence statement. The repository can execute a common preprocessing path, train three model families, produce machine-readable predictions, calculate metrics, and expose them through a local research interface. The audited local run yields the numerical results reported above. It does not establish external generalisation, clinical utility, superiority over published systems, or readiness for deployment.

4.13 Reproducibility check

The reported EfficientNet result was regenerated after the model-comparison run so that the primary metrics, confusion matrix, ROC curve, and per-image predictions refer to one checkpoint and one evaluation command. This matters because

overlapping benchmark processes can otherwise leave a directory containing plots from one model and a metrics file from another. The final artefact set should always be inspected as a group before a value is copied into a table.

The current result is also intentionally modest in its wording. A 67.74% accuracy on 62 local test images is evidence that the pipeline produced predictions above a chance-like baseline in this particular run; it is not evidence that the model will retain the same behaviour on new hospitals, devices, or patients. The confidence intervals and the audit caveat are therefore part of the result, not optional footnotes.

4.14 Relationship to the original project narrative

The submission-facing report removes historical values that could not be traced to a stored, audited experiment. This revision may appear less conclusive than an earlier narrative that described very high accuracy or simulated robustness curves. The change is deliberate. A dissertation should distinguish measured outcomes, planned experiments, and illustrative examples. Keeping those categories separate protects the reader and gives future work a clear starting point.

4.15 Interpreting the custom CNN result

The custom CNN achieved 64.52% accuracy, macro F1 of 0.6452, and ROC-AUC of 0.6708, compared with 56.45%, 0.5374, and 0.6364 for ResNet-50. In a small dataset, a simpler network can sometimes be easier to optimise and less prone to fitting incidental detail than a deeper transfer-learning model. That is a plausible explanation for this result, not a conclusion established by the current experiment. The accuracy intervals overlap: 0.5323–0.7581 for the custom CNN and 0.4355–0.6935 for ResNet-50. The data therefore support the statement that the custom CNN was the better-performing model in this run, while leaving the population-level ranking unresolved.

The result is still scientifically useful because it challenges the assumption that a larger or deeper model must win. It provides a concrete hypothesis for the next study: compare capacity, freezing strategy, and optimisation schedule across several seeds while holding the subject partition and tuning budget fixed. If the

custom CNN remains competitive on a larger external cohort, that would be a meaningful engineering finding. It would still require clinical validation before any medical claim.

Chapter 5

Discussion and Critical Reflection

5.1 Interpretation of the current work

The project demonstrates an end-to-end research implementation, not a validated clinical classifier. Its technical contribution is the consolidation of preprocessing, model loading, inference, and evaluation into one auditable path. This reduces a common source of irreproducibility: reporting metrics from a pipeline that differs from the deployed preprocessing path.

5.2 Data validity and threats to inference

The repaired OASBUD and BrEaST cohort has no cross-partition subject identifiers, but it remains a small local cohort. BUSI source identifiers were discarded during file renaming and that dataset is excluded from the validated run. The checkpoint result in Chapter 4 must therefore not be interpreted as clinical sensitivity, specificity, external generalisation, or comparative superiority.

5.3 Preprocessing claims

Reflection padding preserves the aspect ratio of the image crop by construction, and CLAHE is implemented as an explicit configurable operation. Neither fact establishes improved diagnostic performance. Such a claim requires matched experiments in which crop strategy and CLAHE are varied while data split, model, seed, and training budget are controlled. The same caution applies to synthetic noise simulations: they are useful visual demonstrations, but require measured prediction-level experiments before they become robustness findings.

5.4 Clinical, ethical, and regulatory boundaries

The dashboard is a research interface only. Its predictions, heatmaps, morphology heuristics, and image-quality estimates are not clinical assessments and should not influence care. The interface no longer assigns BI-RADS categories or clinical priority. Any future clinical translation would require a locked model, external validation at subject and site level, calibration and human-factors studies, governance for dataset consent and privacy, and an appropriate medical-device regulatory pathway. A confidence score is not an estimate of diagnostic certainty.

5.5 Remedial study design

The current BrEaST and OASBUD folders have been rebuilt and audited by subject. A later study should preserve the frozen manifest with original identifiers, source dataset, class, image or view, split, and checksum. All architecture, threshold, and preprocessing choices should then be made using training and validation data only. External evaluation should extend the present confidence, calibration, source, and failure-case analyses while recording every exclusion.

5.6 Threats to validity

Several threats remain even after the split audit. The cohort is assembled from public datasets with different acquisition conditions, label conventions, and mask availability. Combining them can increase diversity, but it can also create a dataset fingerprint that a model uses as a shortcut. A site-aware analysis would be needed to determine whether performance is stable when the source dataset changes. The current test partition is too small to support that conclusion.

Label noise is another concern. Folder labels are treated as reference annotations, yet this dissertation has no access to pathology reports, adjudication records, or the circumstances under which each label was assigned. A mislabelled image affects both training and evaluation. Repeating the experiment cannot remove this uncertainty; it can only estimate behaviour relative to the supplied labels. A future study should document the label source and, where feasible, include independent review.

The preprocessing path can introduce selection effects. Mask-guided crops are available only for images with recognised paired masks, while the fallback crop is used when a mask is missing. If mask availability correlates with dataset source or class, the model may see systematically different input distributions. The audit should therefore report mask coverage by split and class, and an ablation should compare masked and unmasked policies under a common manifest.

5.7 Failure analysis as a research activity

Failure analysis should be planned before the final test run, not added as an anecdotal gallery afterwards. A useful record contains the image identifier, source dataset, subject group, true class, predicted class, probability, preprocessing configuration, mask status, and a short reviewer note. Sorting by uncertainty identifies borderline cases; sorting by confidence identifies potentially dangerous errors; grouping by source identifies possible domain shift. These views complement the aggregate confusion matrix.

The reviewer should avoid treating a heatmap as an explanation by default. The same image can be transformed through direct resize, centre crop, and square padding to test whether the prediction changes with geometry. If a small border alteration changes the class, the model may be sensitive to preprocessing rather than stable lesion evidence. If brightness or contrast perturbations change a high-confidence prediction, the case belongs in a robustness report. The outcome may be a model improvement, a revised limitation, or a decision to drop a transformation.

5.8 Reproducibility and maintainability

The project is stronger when the report, notebook, API, and dashboard all read from the same generated artefacts. A manually copied metric is a maintenance risk: it can remain plausible while no longer matching the checkpoint. The current arrangement uses the evaluator as the source of truth for predictions, tables, and figures. New charts should be generated from the CSV or JSON output, and new prose should identify the run that produced a number.

Maintainability also depends on preserving a small interface between modules.

Dataset discovery should not know about HTML; the evaluator should not silently change label mapping; and the dashboard should not reimplement preprocessing in JavaScript. This separation makes review easier and reduces the chance that a visual enhancement changes the experiment. It also supports a clean handover to a supervisor or another student, who can run the audit and inspect generated files before trusting the result.

5.9 Clinical translation barriers

Moving from an academic prototype to clinical research would require more than a larger model. The intended use, target population, acquisition protocol, and acceptable error trade-offs would need to be specified. External validation should include new subjects and, ideally, new sites or devices. Calibration should be checked in the intended population, and the interface should be studied with representative clinicians to identify automation bias, alert fatigue, and misunderstanding of uncertainty.

Governance is equally important. Images and subject identifiers must be handled under relevant consent, privacy, and data-sharing agreements. A model update requires versioning and change control. Any clinical claim would need an appropriate regulatory and institutional review pathway. The current local dashboard deliberately stops short of these functions: it is an educational and research demonstration, not a clinical decision-support product.

5.10 Practical recommendations

The next iteration should prioritise four actions. First, keep the local BUSI derivative outside all validated claims unless a separately acquired original release can be curated from a versioned manifest. Second, retain the frozen BrEaST and OASBUD subject manifest before further tuning. Third, repeat the architecture and preprocessing comparison across predeclared seeds, with any calibration fitted on validation data. Fourth, regenerate the dissertation, README, notebook, presentation, and dashboard from a run-specific output directory. These actions address the largest sources of uncertainty without requiring a more complex neural architecture.

5.11 What a successful next experiment would look like

A successful next experiment would begin with an explicit protocol signed off before training. It would define the target label, the eligible subjects, the grouping rule, the mask policy, the preprocessing candidates, the tuning budget, and the primary metrics. The test manifest would then be frozen and stored separately from development artefacts. Model selection would use only the training and validation partitions, with each seed and configuration recorded.

The present report shows both aggregate and case-level evidence. A larger external report should retain the same structure: aggregate tables with confidence intervals and calibration, plus case-level files that let a reviewer trace errors to source images. A qualitative review could then identify recurring artefacts or annotation problems. The review would generate hypotheses, not justify deleting difficult cases after their predictions have been inspected.

5.12 Value of a negative or inconclusive result

If the corrected experiment shows no meaningful difference between preprocessing choices, that is a useful finding. It would suggest that the visual preference for one transformation is not supported by the measured task under the chosen conditions. If all models perform poorly on an external cohort, that would identify domain shift rather than invalidate the engineering work. A transparent failure can guide data collection, annotation, and study design more effectively than a confident but fragile success.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This dissertation delivers a reproducible research framework for breast-ultrasound image classification with shared preprocessing, multiple convolutional architectures, transparent evaluation artifacts, and a read-only dataset audit. The work establishes that the software can train, evaluate, and serve a research demonstration, while also exposing why reproducibility controls are necessary before reporting biomedical performance.

The repaired OASBUD and BrEaST dataset copy supports a subject-level local evaluation, but the missing BUSI provenance and limited cohort size prevent external or clinical claims. The fresh checkpoint result is a local research finding only. Earlier static performance, pseudo-labelling, and noise-resilience values have therefore been withdrawn as empirical conclusions.

6.2 Future work

The BrEaST and OASBUD cohort now retains subject grouping and passes the local split audit. The immediate priority for a later experiment is to freeze that manifest, keep the untraceable BUSI derivative excluded, and save the complete training history for each predeclared seed. Model selection should use validation data, followed by one evaluation on an untouched test partition and regeneration of every table and figure from saved predictions. External-site evaluation, fitted calibration, expanded error analysis, and prospective human-factors validation are prerequisites for any clinical-development claim.

Future technical work may compare self-supervised pretraining, attention architectures, and domain adaptation, but only after the basic split and provenance controls are in place. The appropriate success criterion is not a headline accuracy

alone; it is a transparent, repeatable estimate whose limitations are clear to researchers, clinicians, and patients.

6.3 Closing perspective

The project shows that a classifier cannot be interpreted in isolation. Its meaning depends on the files used to train it, the subject split, the transformation applied before inference, the display threshold, and the evidence retained after evaluation. Improving one component while leaving the others undocumented produces a more polished system without resolving the central uncertainty.

The repaired MMU-formatted project makes that chain visible. A reader can follow the work from literature and dataset audit through preprocessing, model construction, evaluation, and web presentation. The current numerical result is modest and uncertain, but it is traceable. That is a suitable foundation for a dissertation because it permits a supervisor, examiner, or future researcher to identify exactly what should be changed in the next experiment.

The recommended next milestone is not a claim of clinical readiness. It is a locked, subject-independent, multi-seed evaluation with restored provenance and an external test cohort. If that study supports improvement, the evidence will justify a stronger conclusion. If it does not, the project will still have produced a clear account of the conditions under which the model fails. Both outcomes are more valuable than an unreproducible headline number.

6.4 Final submission checklist

Before submission, the author should confirm that the ethics identifier and declaration are complete, that the university's required word-count rules have been applied, and that the PDF opens without missing figures or references. The source archive should contain the MMU template files, the chapter sources, the bibliography, figures, and a short compilation README. Generated auxiliary files may be excluded if the submission portal requests source-only content, but the final PDF and the exact source used to create it should be retained.

The technical checklist should confirm that the audit passes for every dataset included in the headline result, that the checkpoint metadata names the same

preprocessing used by the evaluator, and that the dashboard does not display withdrawn historical values. The narrative checklist should confirm that every numerical claim can be located in a generated artefact and that clinical limitations appear beside the result they qualify.

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Appendix A

Data Audit and Reproducibility Record

A.1 Purpose of this record

This appendix connects the numerical results in Chapter 4 to the files and commands that produced them. It records the cohort used by the validated experiment, the subject separation rule, the stored outputs, and the limits on reproduction. The record is included because a checkpoint alone is not enough to reconstruct an experiment. A reviewer also needs to know which images entered each partition, how labels were mapped, which preprocessing path was active, and which evaluator generated the reported values.

The primary run used the repaired BrEaST and OASBUD folders. Their filenames retain identifiers that can be grouped at subject level. The audit found no subject identifier in more than one of the training, validation, and test partitions. BUSI was omitted from the validated run because the local files had been renamed to generated identifiers. Without the original manifest, the repository cannot establish whether two BUSI images belong to the same patient. This exclusion reduces the available sample size, but it prevents an unverifiable dataset from being presented as independent test evidence.

A.2 Audited cohort snapshot

Table A.1 reproduces the class counts from `outputs/data_audit.json`. The normal BrEaST images are loaded as non-malignant examples because the executable task has two output classes. This mapping is explicit in the dataset loader and should be reviewed before a later clinical study because a normal image and a benign lesion do not express the same clinical condition.

Table A.1: Audited image counts used by the validated experiment.

Dataset	Partition	Benign	Malignant	Normal	Total
BrEaST	Training	101	60	3	164
BrEaST	Validation	19	13	1	33
BrEaST	Test	19	13	0	32
OASBUD	Training	66	72	0	138
OASBUD	Validation	14	16	0	30
OASBUD	Test	14	16	0	30
Combined	Training	167	132	3	302
Combined	Validation	33	29	1	63
Combined	Test	33	29	0	62

The audit uses a dataset-specific identifier rule. For BrEaST, it retains the caseNNN prefix. For OASBUD, it removes the suffix that identifies different views of the same lesion. It then builds a mapping from each derived identifier to every partition in which that identifier appears. An audit passes only when the identifiers are recoverable and each identifier occurs in one partition. The stored report records `audit_passed`: true, lists breast and oasbud as the audited datasets, and contains an empty `cross_split_subjects` object for both.

A.3 Experiment configuration

The validated checkpoint uses EfficientNet-B0 with a two-output classifier. The input pipeline applies CLAHE, uses the padded region-of-interest crop, resizes the result to 224×224 pixels, and normalises the three channels with the ImageNet mean and standard deviation used by the transfer-learning backbone. Training uses batch size 16, an initial learning rate of 10^{-4} , weight decay of 10^{-4} , class-weighted cross-entropy, and seed 42. The maximum epoch budget is 25, with early stopping after five validation epochs without improvement in validation loss.

Evaluation loads the selected checkpoint without downloading new pretrained weights. It applies no random test augmentation and writes one row per test image. The stored primary result contains 62 rows, with 33 benign or non-malignant labels and 29 malignant labels. The confusion matrix is

$$\begin{bmatrix} 21 & 12 \\ 8 & 21 \end{bmatrix}, \quad (\text{A.1})$$

where rows are true classes and columns are predicted classes in the order benign, malignant. These counts yield 42 correct predictions and 20 errors. The corresponding summary is 0.6774 accuracy, 0.6774 macro F1, 0.6803 macro recall, 0.7419 ROC-AUC, and 0.0652 expected calibration error. The evaluator also records bootstrap 95% intervals of 0.5645 to 0.7903 for accuracy, 0.5589 to 0.7889 for macro F1, and 0.6217 to 0.8616 for ROC-AUC.

A.4 Reproduction sequence

The following commands are run from the repository root with the project environment active. They are shown on separate lines so that paths remain readable in the printed report.

```
venv/bin/python scripts/audit_data.py \
  --datasets breast oasbud \
  --output outputs/data_audit.json

venv/bin/python train.py \
  --model efficientnet_b0 \
  --epochs 25 --batch-size 16 --lr 0.0001 --seed 42

venv/bin/python evaluate.py \
  --checkpoint outputs/efficientnet_b0_model.pth \
  --split test

venv/bin/python scripts/benchmark_models.py \
  --split test \
  --models efficientnet_b0 resnet50 custom_cnn

venv/bin/python scripts/generate_report_figures.py
```

Training invokes the audit before the first optimisation step. If the selected dataset fails, the command stops unless the author explicitly supplies

--allow-unaudited-data. That override marks an engineering run; it must not be used to support the validated headline result. The evaluation step reloads the model architecture named inside the checkpoint, loads its stored state dictionary, uses the configured datasets, and writes the result bundle described below.

A.5 Stored evidence bundle

Table A.2 lists the files required to trace the reported result. The files should be retained together because replacing only one item can leave plots, predictions, and summary values referring to different runs.

Table A.2: Primary reproducibility artefacts retained by the project.

Artefact	Role in the evidence chain
outputs/data_audit.json	Dataset selection, partition counts, identifier status, audit outcome, and exclusions.
outputs/efficientnet_b0_model.pth	EfficientNet-B0 checkpoint evaluated for the primary result.
outputs/predictions.csv	Test-image path, true label, predicted label, and malignant-class probability.
outputs/metrics.json	Aggregate metrics, confusion matrix, uncertainty intervals, preprocessing configuration, generation time, and audit checksum.
outputs/model_benchmark.json	Same-split comparison of EfficientNet-B0, ResNet-50, and the custom CNN.
outputs/extended_evaluation.json	Average precision, Brier score, class-level results, and source-stratified results derived from the stored primary predictions.
outputs/confusion_matrix.png	Matrix generated from the stored test predictions.
outputs/roc_curve.png	ROC curve generated from malignant-class probabilities.
scripts/generate_report_figures.py	Rebuilds dataset composition, architecture comparison, uncertainty, dataset-quality, and image-level prediction figures from local evidence.
figures/fig11_web_interface.png	Browser capture of the implemented dashboard state reproduced in Figure 3.3.
figures/fig14_web_interface_tour.png	Six-screen browser capture showing the implemented research workflow and use boundaries in Figure 3.4.

The metrics record includes a SHA-256 checksum for the audit file associated with the result:

```
0885667e79427e5ec431c573a11bf0d28
b8cb0770fbe2e9214ce27dacc5bccb9
```

This checksum links the summary to one audit output. It does not replace checksums for the source images, checkpoint, environment, or repository revision. A future release should store those values in a run-specific directory so that later experiments cannot overwrite the evidence used in the dissertation.

A.6 Independent checks and remaining limits

The automated test suite checks six software contracts: every crop strategy returns the configured model dimensions, mask discovery recognises the supported suffixes, an unaudited result is labelled as provisional, the API does not recommend clinical action, the primary prediction record excludes BUSI, and the extended evaluation agrees with the stored sample count and confusion matrix. These tests guard against accidental software changes. They do not measure diagnostic safety or external validity.

Reproduction also depends on the software environment and hardware. The seed fixes the intended random streams in Python, NumPy, PyTorch, and CUDA where available, but library versions and non-deterministic kernels can still change the fitted weights. A faithful rerun should therefore record the operating system, Python version, PyTorch and torchvision versions, device type, dependency lock, Git revision, and checkpoint checksum. The current appendix provides the executable path and stored result; it does not claim bitwise identity on every machine.

The test sample is small and comes from two local public-data preparations. No independent hospital cohort, prospective reader study, or clinical workflow evaluation was performed. The metrics describe this audited local partition only. The wide bootstrap intervals and overlapping architecture intervals should remain visible whenever the result is presented.

A.7 Post-hoc evaluation record

The additional precision-recall, reliability, and source-stratified analyses use only the 62 rows in `outputs/predictions.csv`. The malignant class is the positive class and the stored classification uses a threshold of 0.5. No observation was removed, relabelled, or reweighted for these plots. The analysis writes `outputs/extended_evaluation.json` so the values printed in Chapter 4 can be checked without reading values from the image.

The source check identifies 32 BrEaST images and 30 OASBUD images from their stored relative paths. It is exploratory because each subset is small and the two sources differ in acquisition and reconstruction. The calculation is still useful: it shows that the pooled result combines different error profiles. Any later

external study should retain the source field explicitly rather than recovering it from a path.

The old checkpoints do not contain per-epoch histories. Their selected epochs and validation summaries are reported, but a full loss or accuracy curve would require information that was not saved. The revised training code stores that history for future experiments. It does not retroactively create evidence for the completed runs.

Appendix B

Software Implementation and Safety Boundary

B.1 System purpose

The repository contains a FastAPI service and a browser dashboard for inspecting the dissertation pipeline. The interface accepts a research image, applies the shared preprocessing function, loads a verified checkpoint, and returns the class probabilities with supporting visual and descriptive outputs. It also exposes the stored benchmark and data-audit files used by the report. This makes the implementation easier to inspect during assessment because the evaluator, API, notebook, and figures read from the same evidence bundle.

The application is a research demonstration. It is not registered or evaluated as a medical device, and it does not provide a diagnosis, BI-RADS category, treatment recommendation, referral decision, or clinical priority. A displayed benign or malignant class is the output of the fitted two-class model under the selected preprocessing configuration. The meaning is limited to the labels used during training.

B.2 Component map

Table B.1 identifies the main implementation files. The separation is intentional: data preparation, model construction, evaluation, and presentation can be inspected without treating the browser interface as the source of experimental truth.

Table B.1: Principal software components.

Component	Responsibility
src/dataset.py	Dataset discovery, mask pairing, CLAHE, crop strategy, square padding, resizing, augmentation, and dataloader construction.
src/models.py	Factory for EfficientNet-B0, ResNet-50, and the custom four-stage CNN.
src/train.py	Audit gate, reproducible seed, optimisation loop, validation, early stopping, and checkpoint metadata.
src/evaluate.py	Locked-checkpoint evaluation, per-image predictions, aggregate metrics, confidence intervals, calibration estimate, confusion matrix, and ROC curve.
src/api.py	Upload validation, checkpoint registry, single and batch prediction, Grad-CAM, descriptive image heuristics, evidence routes, and research-only response wording.
web/	Static dashboard, evidence views, comparison controls, and explanatory text.
tests/test_research_pipeline.py	Regression tests for preprocessing dimensions, masks, evidence status, and safety wording.

B.3 Inference request path

A single-image request follows a defined sequence. First, the API verifies the extension, confirms that the upload is non-empty, and enforces the 20 MB size limit. It decodes the image and rejects unreadable files. A second gate checks minimum dimensions, excessive colour-channel spread, mean saturation, near-white document-like appearance, very low luminance, and insufficient intensity variation. These are simple input heuristics. They can reject obvious colour photographs or blank files, but they cannot certify that an accepted image is a breast ultrasound scan.

The accepted image enters `preprocess_image`, which is also used by dataset evaluation and command-line prediction. The caller selects CLAHE and one of three geometric strategies: padded region-of-interest crop, centre crop, or direct resize. The processed image is converted to a tensor and normalised.

The API then selects a checkpoint only when the checkpoint metadata names a supported architecture. It reconstructs that architecture, loads the stored parameters, switches to evaluation mode, and computes a softmax probability vector.

The display layer compares the largest softmax score with the configured abstention threshold of 0.60. Scores below that threshold are surfaced as UNCERTAIN. This changes how the interface communicates the result; it does not create a third trained class and does not calibrate the probability. The API therefore returns the raw probabilities and the abstention state separately.

Grad-CAM is calculated from a late convolutional layer selected for each architecture. The resulting heatmap is resized and overlaid on the processed image. The overlay shows which feature-map regions were influential for that forward and backward pass. It is not a lesion segmentation, causal explanation, or validation that the model used medically appropriate evidence. The interface treats it as an inspection aid.

B.4 API surface

Table B.2 summarises the routes used by the dashboard. Route names are included to make the implementation demonstrable without copying large code listings into the dissertation.

Table B.2: Research API routes exposed by the implementation.

Route	Method	Purpose
/health	GET	Reports device, class names, checkpoint availability, and service state.
/predict	POST	Runs one accepted image through a selected verified model and preprocessing configuration.
/predict/compare	POST	Compares the available trained architectures on the same uploaded image.
/predict/batch	POST	Processes a controlled batch through the shared inference path.
/api/benchmarks	GET	Returns the stored internal architecture benchmark.
/api/data-audit	GET	Returns the stored data-audit record.
/api/evidence/...	GET	Serves allow-listed evidence files used by the dashboard.
/api/dataset/cohort	GET	Selects a balanced demonstration cohort from an available dataset split.
/api/noise/simulate	POST	Applies a seeded synthetic perturbation for research inspection.

The benchmark route reads `outputs/model_benchmark.json`; it does not contain a second set of handwritten scores. The data-audit route follows the same rule. This reduces the chance that the website displays values that differ from the submitted report. A future version should add authentication and a stricter origin policy before the service is exposed outside a local research machine. The current permissive CORS setting supports local development and is not a production security configuration.

B.5 Image descriptors and uncertainty

The interface calculates descriptive texture and geometry proxies to support inspection. These include local intensity variation, normalised Laplacian variance, the proportion of pixels that differ strongly from a median-filtered reference, an estimated signal-to-noise ratio, central dark-region localisation, and simple intensity ratios around the selected region. The names shown to the user describe

measured image statistics. They do not identify scanner faults, pathology, lesion margins, posterior acoustic behaviour, or other clinical findings.

The distinction matters because a deterministic formula can still have an invalid interpretation. For example, the localisation routine searches a central region for a dark structure after smoothing and thresholding. When it cannot find a clear peak, it falls back to the image centre. That behaviour keeps the demonstration executable, but it is not a validated lesion detector. The research report returned by the API therefore labels these outputs as heuristics and states that no clinical action may be inferred.

B.6 Verification checklist

The following checks are appropriate before demonstrating or submitting the implementation:

1. Run the subject-level audit and confirm that the selected datasets pass.
2. Confirm that the checkpoint architecture matches the metadata and that no random-weight fallback is exposed.
3. Run the automated tests and retain the output with the project record.
4. Start the local service and check that `/health` reports a ready model.
5. Submit one valid research image and confirm that preprocessing settings, probabilities, abstention state, and model name are returned.
6. Submit an empty file, an unsupported extension, and an obvious colour photograph to verify controlled rejection.
7. Compare the dashboard values with the generated model benchmark JSON file.
8. Confirm that the interface contains the research-use warning and gives no diagnosis or treatment advice.

At the time of this report, six automated tests pass. They cover important preprocessing, evidence-boundary, and stored-result contracts but do not exercise every API route or browser interaction. A stronger verification

suite would add endpoint tests, corrupted-checkpoint tests, file-size boundary cases, malformed-image inputs, deterministic preprocessing snapshots, and an end-to-end browser test that compares displayed metrics with the JSON evidence.

B.7 Known limitations and recommended development

The domain gate is heuristic and can produce both false acceptance and false rejection. A grayscale image from another modality may pass, while a valid ultrasound frame containing colour Doppler or annotations may fail. A dedicated modality and anatomy classifier would require its own labelled training data, independent evaluation, and failure analysis. Until that work is completed, the gate should be described as an obvious-input filter.

The current dashboard runs locally and loads checkpoints from disk. It has no user authentication, audit trail for individual requests, encrypted clinical storage, retention policy, or role-based access control. Those omissions are acceptable for a local demonstration using permitted research data, but they preclude clinical deployment. Any move beyond the dissertation environment would require information-governance review, security testing, monitoring, version control for models and preprocessing, and an approved process for reporting failures.

The next scientific step is separate from interface polish. The models should be trained across predeclared seeds, tuned with equal budgets, and evaluated on a larger external subject cohort. Threshold selection and calibration must use validation data only. A later reader study could examine whether the interface helps or distracts trained users, but such a study would require an approved protocol and should not be inferred from the present software demonstration.

Appendix C

Terms of Reference

This appendix reproduces the project Terms of Reference in the university format. The project title and module code have been updated to match the final dissertation and the current 6G7V0007 assignment brief. The remaining content records the project plan agreed at the start of the work, including the original proposed dataset and evaluation scope.

7V0007 MSc Project
NPC ToR Coversheet

Department of Computing and Mathematics Computing and Digital Technology Postgraduate Programmes Terms of Reference Coversheet

Student name:	Akerele David Damilola
University I.D.:	25908322
Academic supervisor:	Dr Moi Hoon Yap
External collaborator (optional):	
Project title:	Breast Cancer Ultrasound Classification Using Machine Learning
Degree title:	MSc Artificial Intelligence
Project unit code:	6G7V0007
Credit rating:	60
Start date:	01/05/2026
ToR date:	06/06/2026
Intended submission date:	11/09/2026
Signature and date student:	Akerele David Damilola 06/06/2026
Signature and date external collaborator (if involved):	

This sheet should be attached to the front of the completed ToR and uploaded with it to Moodle.

Manchester Metropolitan University

MSc Artificial Intelligence

2025 / 2026

Project Terms of Reference

Breast Cancer Ultrasound Classification Using Machine Learning

Aims & Objectives

Breast cancer is one of the most prevalent and life-threatening cancers worldwide. Early and accurate diagnosis is critical to improving patient survival rates, and medical imaging — particularly ultrasound — is a widely used, non-invasive diagnostic tool. However, the manual interpretation of ultrasound images is time-consuming and subject to inter-observer variability. This project aims to design, develop, and evaluate machine learning models capable of automatically classifying breast cancer ultrasound images into benign, malignant, and normal categories.

The overarching aim is to contribute to the development of an AI-assisted diagnostic tool that can support clinicians in making faster, more consistent, and more accurate breast cancer assessments from ultrasound imagery.

To achieve this aim, the following objectives have been identified:

- Terms of Reference — Define the project scope, aims, objectives, evaluation plan, and activity schedule.
- Literature Review — Conduct a comprehensive survey of existing breast cancer detection methods, machine learning and deep learning architectures, and publicly available ultrasound datasets.
- Data Acquisition & Pre-processing — Identify, obtain, and prepare a suitable labelled ultrasound image dataset (e.g. BUSI dataset), including image normalisation, augmentation, and train/validation/test splitting.
- Model Design & Development — Implement and compare multiple classification approaches, including traditional machine learning classifiers (e.g. SVM, Random Forest) applied to hand-crafted features, and convolutional neural network (CNN) architectures (e.g. ResNet, VGG, EfficientNet) applied to raw image data.
- Evaluation & Testing — Assess model performance using standard metrics: accuracy, precision, recall, F1-score, and AUC-ROC; compare approaches and identify strengths and limitations.
- Final Report — Document all stages of the project, present findings, and critically reflect on outcomes and future directions.

The central research hypothesis guiding the project is:

"Machine learning models, and in particular deep convolutional neural networks, can achieve clinically meaningful classification accuracy on breast cancer ultrasound images, outperforming traditional feature-based approaches."

Learning Outcomes

In accordance with the MSc Artificial Intelligence degree learning outcomes (as specified in the 6G7V0007 Project Handbook, Appendix D), the following outcomes will be addressed and demonstrated through this project:

- Choose, customise, and combine suitable core techniques for solving a given AI problem — the project will evaluate and combine CNN architectures and classical ML pipelines to address the image classification task.
 - Making descriptive and inferential statements about data via appropriate processing techniques and models, accounting for aspects such as uncertainty, data quality, and model fitness — the project will involve statistical analysis of model outputs, uncertainty estimation, and performance benchmarking.
 - Process and analyse, effectively and efficiently, data of varying scales and from heterogeneous sources, formats, and systems, using a range of suitable languages, tools, and environments — the project will involve handling raw medical image data, applying augmentation pipelines, and working across Python, TensorFlow/Keras, and scikit-learn.
 - Build AI solutions with good software practices such as in code reuse, separation of concerns, modularity, testing, and documentation — all code will be structured, version-controlled (GitHub), and accompanied by clear documentation.
-

Project Description

Breast cancer ultrasound imaging produces greyscale images of breast tissue; lesions appear as hypoechoic regions and must be classified by their morphological characteristics. Manual classification is challenging and error-prone, making it a strong candidate for AI-assisted analysis.

This project will primarily utilise the Breast Ultrasound Images (BUSI) dataset, a publicly available dataset containing 780 images across three classes: normal, benign, and malignant. The dataset includes segmentation masks alongside images, enabling optional use of region-of-interest (ROI) extraction as a pre-processing step.

The project workflow will proceed through the following stages:

- Data pre-processing: image resizing, pixel normalisation, histogram equalisation, and data augmentation (flipping, rotation, zoom) to address class imbalance and improve model generalisation.
- Feature extraction (for classical ML): texture features using Grey-Level Co-occurrence Matrix (GLCM), shape features, and histogram of oriented gradients (HOG).
- Classical ML pipeline: SVM, Random Forest, and k-NN classifiers trained on extracted features; hyperparameter tuning via cross-validation.
- Deep learning pipeline: transfer learning with pre-trained CNN backbones (ResNet50, VGG16, EfficientNetB0) fine-tuned on the BUSI dataset; custom CNN architecture as a baseline.
- Comparative evaluation: systematic comparison of all approaches using held-out test data; analysis of confusion matrices, ROC curves, and learning curves.

Legal, social, ethical and professional considerations: The BUSI dataset is publicly available for academic research and does not contain personally identifiable information. No human participants will be recruited for this project. The project will comply fully with Manchester Metropolitan University's Academic Ethical Framework. Ethical approval will be obtained via the EthOS system prior to data collection. Findings will be reported honestly and limitations clearly acknowledged, in line with professional computing ethics standards.

References

The following initial references will be used to begin the literature survey:

- Al-Dhabyani, W., Gomaa, M., Khaled, H. and Fahmy, A. (2020) 'Dataset of breast ultrasound images'. Data in Brief, 28, p.104863. [BUSI dataset]
- Shen, L., Margolies, L.R., Rothstein, J.H., Fluder, E., McBride, R. and Sieh, W. (2019) 'Deep learning to improve breast cancer detection on screening mammography'. Scientific Reports, 9(1), p.12495.
- Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., van der Laak, J.A., van Ginneken, B. and Sánchez, C.I. (2017) 'A survey on deep learning in medical image analysis'. Medical Image Analysis, 42, pp.60–88.
- He, K., Zhang, X., Ren, S. and Sun, J. (2016) 'Deep residual learning for image recognition'. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp.770–778.
- Tan, M. and Le, Q.V. (2019) 'EfficientNet: Rethinking model scaling for convolutional neural networks'. Proceedings of the 36th International Conference on Machine Learning (ICML), PMLR 97, pp.6105–6114.
- Simonyan, K. and Zisserman, A. (2014) 'Very deep convolutional networks for large-scale image recognition'. arXiv preprint arXiv:1409.1556.
- Cheng, J.Z., Ni, D., Chou, Y.H., Qin, J., Tiu, C.M., Chang, Y.C., Huang, C.S., Shen, D. and Chen, C.M. (2016) 'Computer-aided diagnosis with deep learning architecture: applications to breast lesions in US images and pulmonary nodules in CT scans'. Scientific Reports, 6(1), p.24454.

Evaluation Plan

Evaluation will be conducted at two levels: technical and process-oriented.

Technical evaluation will assess the performance of all developed models against a held-out test set (20% of available data). Metrics will include:

- Overall classification accuracy
- Per-class precision, recall, and F1-score
- Area under the ROC curve (AUC-ROC) for multi-class discrimination
- Confusion matrix analysis to identify misclassification patterns

Model performance will be compared against published benchmarks on the BUSI dataset and against each other to determine whether deep learning approaches offer meaningful advantages over classical ML methods. Ablation studies will be conducted to assess the contribution of individual components (e.g. data augmentation, transfer learning, fine-tuning).

Process evaluation will compare actual progress against the activity schedule, reflecting on deviations and how they were managed. The project will also be self-evaluated against the initial research hypothesis and degree learning outcomes, and suggestions for future work will be made in the final report.

Activity Schedule

The project is planned across the period May 2026 to September 2026. The following table and Gantt chart summarise the key tasks, start and end dates, and durations:

Task	Start Date	End Date	Duration (Days)
Terms of Reference & Ethics	01/05/2026	06/06/2026	36

Literature Review	01/05/2026	15/07/2026	75
Data Acquisition & Pre-processing	07/06/2026	05/07/2026	28
Model Design & Development	06/07/2026	09/08/2026	34
Evaluation & Testing	10/08/2026	24/08/2026	14
Report Writing	20/08/2026	11/09/2026	22

Note: Literature review and data pre-processing tasks overlap intentionally to allow iterative refinement of the approach as relevant papers are identified. Report writing begins before evaluation is complete to allow early drafting of background and methodology chapters.

Signing and Dating

Submission of this document to Moodle by the student counts as the student signature. Acceptance mark on Moodle by the supervisor counts as the supervisor signature, in accordance with the 6G7V0007 Project Handbook.

Student: Akerele David Damilola Date: 06/06/2026

Supervisor: Dr Moi Hoon Yap Date: _____

Appendix D

EthOS Ethics Documentation

The EthOS approval email is reproduced first in this appendix. Dated 18 June 2026, it records a favourable ethical opinion for the project and gives EthOS Reference Number 92016. It states that the approval is in place until six months after the end date recorded in the application documentation, 11 September 2026.

The dissertation uses the registered project title “Breast Cancer Ultrasound Classification Using Machine Learning”. The signed EthOS application record follows for completeness and identifies David Akerele, student ID 25908322, supervisor Dr Moi Hoon Yap, and the student and supervisor signatures dated 18 June 2026.



📧 donotreply@infonetica.net <donotreply@infonetica.net>

Thursday, 18 June 2026 at 11:19 am

To: 🟢 David Akerele; Cc: 🟡 Moi Hoon Yap ▾

This email originated from outside of Manchester Met. Do not click links or open attachments unless you recognise the sender and believe the content to be safe. Please contact the IT ServiceDesk if you have any concerns, <https://www.mmu.ac.uk/about-us/professional-services/itd/about/contact>



18/06/2026

Project Title: Breast Cancer Ultrasound Classification Using Machine Learning

EthOS Reference Number: 92016

Ethical Opinion

Dear David Akerele,

The above application was reviewed by and on the 18/06/2026, was given a favourable ethical opinion. The approval is in place until six months after the end date recorded in your application documentation (11/09/2026).

Approved Documents

Conditions of favourable ethical opinion

The favourable ethical opinion is granted with the following conditions

Approval is in place for your UG/PGT project

This approval is only valid for Undergraduate (UG) and Post Graduate Taught (PGT) projects and does not grant approval for any Staff or PGR projects.

Adherence to Manchester Metropolitan University's Policies and procedures

This ethical approval is conditional on adherence to Manchester Metropolitan University's Policies, Procedures, guidance and Standard Operating procedures. These can be found on the Manchester Metropolitan University Research Ethics and Governance webpages.

Amendments

If you wish to make a change to this approved application, you will be required to submit an amendment. Please visit the Manchester Metropolitan University Research Ethics and Governance webpages or contact your Faculty research officer for advice around how to do this.

We wish you every success with your project.

Science and Engineering Research Ethics and Governance Committee

2.1.06

For help with this application, please first contact your Faculty Research Officer. Their details can be found [here](#)

Introduction

Thank you for visiting the Ethics Online System (EthOS)

All members of staff and students who are about to undertake a research project or investigation are required to use EthOS to determine what form of ethical approval, if any, is required.

The questions within EthOS will be automatically generated based on your previous answers so you may not see all sections.

Once you have submitted your application then it will receive a proportionate level of review based on your answers.

If, during review, any changes are required then the application will be returned to you. Once you've made the changes then you will be able to resubmit.

Please be aware that 'submission' does **not** constitute approval - you will need to wait until an approval letter has been issued (via email) by the system.

Applicant Information

Title

First Name

Surname

Email

A2 Manchester Metropolitan University ID number

Applicant Status

B1 Are you completing this application as a...

UG/PGT Student - Default application

B2 Please select the faculty that is most closely aligned with your project

Science & Engineering

B6 Please select the Science and Engineering department that is responsible for this project

School of Computing and Mathematics

B3.1 If you have selected 'Other' as your department, please specify

B1.2 Manchester Metropolitan Supervisor Details

First Name

Moi Hoon

Surname

Yap

Email

m.yap@mmu.ac.uk

Please click "Next" or "Navigate" on the sidebar to proceed with the application.

Project Details

D0.0 Please enter the module code for your course. This can be found on Moodle. If you are unsure, please contact your supervisor.

6G7V0007_2526_9F

D1 Full Project Title

Breast Cancer Ultrasound Classification Using Machine Learning

D2 Project Short Title

This should be the title shown in your Participant Information Sheet (PIS), consent form, advertising materials and any other supporting documents.

Breast Cancer Ultrasound Classification Using Machine Learning

D3 Estimated Start Date

29/05/2026

The start of your data collection must be in the future. It is not the date when you registered/started at the university. When selecting this date please allow time for your application to be reviewed.

D4 Project End Date

This should be the date when the final report for the study is submitted. If you are unsure of this date, please put the date when all aspects of the project will be complete.

For students, this is the final submission date of your thesis/dissertation/assignment.

11/09/2026

Please click "Next" or "Navigate" on the sidebar to proceed with the application.

UG/PGT Regulatory Screener

Begin UGPGT screening section

This section will ask a series of questions that will determine if the 'UG/PGT - Default Application' is the correct pathway to approval for you. Please answer these questions carefully.

If you are unsure how to answer the following questions then please contact your supervisor in the first instance. If you and your supervisor are having difficulty then please contact your [Faculty Ethics](#) team.

☒ Begin

UGPGT0 Does your project contain any of the following:

- Approval from an external regulatory body (including, but not limited to: NHS (HRA), HMPPS, MOD etc.);
- The use of ionising radiation or exposure to radioactive materials;
- Misleading participants;
- Research without the recruited participants' consent;
- The use of novel techniques, even where apparently non-invasive, whose safety may be open to question;
- Engaging in, witnessing, or monitoring criminal activity;
- Engaging with, or accessing terrorism related materials;
- A requirement for security clearance or licence to access participants, data or materials;
- Animals, animal tissue, new or existing human tissue, or biological toxins and agents;

☐ Yes

☐ Yes but I have discussed this question with my supervisor and have been advised to continue

☒ No

UGPGT1

Does your project contain any of the following:

- Processing of data relating to criminal convictions and offences
- Large Scale processing of sensitive special category data or data of a highly personal nature
- Processing of data concerning vulnerable individuals
- Use of MRI images or MRI data
- Systematic monitoring of a publicly accessible place on a large scale
- Processing involving the use of innovative technologies, or the novel application of existing technologies (such as Artificial Intelligence, Internet of Things (IoT), or location tracking)
- Processing of biometric or genetic data
- Matching or combining data sets from more than one source
- The processing of personal data which could result in a risk of harm to the data subject in the event of a security breach
- The processing of personal data without providing a privacy notice or participant information sheet directly to the individual

- ☐ Yes
- ☐ Yes but I have discussed this question with my supervisor and have been advised to continue
- ☒ No

UGPGT2

Does your project contain any of the following:

- Travel to countries considered "High Risk" on the Government's [Foreign Travel Advice \(FCDO\) website](#);
- Working with Hepatitis, Human T-Cell Lymphotropic Virus Type iii (HTLV iii), or Lymphadenopathy Associated Virus (LAV) or the mutants, derivatives or variations thereof, or Acquired Immune Deficiency Syndrome (AIDS), or any syndrome or condition of a similar kind;
- Working with Transmissible Spongiform Encephalopathy (TSE), Creutzfeldt-Jakob Disease (CJD), variant Creutzfeldt-Jakob Disease (vCJD) or new variant Creutzfeldt-Jakob Disease (nvCJD);
- Working in hazardous areas or high risk countries;
- Working with hazardous substances outside of a controlled environment;
- Working with persons with a history of violence, substance abuse or a criminal record.

- ☐ Yes
- ☐ Yes but I have discussed this question with my supervisor and have been advised to continue
- ☒ No

UGPGT3

Does your project contain any of the following:

- Physically invasive techniques;
- Ingestion of food, liquids, or drugs;
- Physical testing;
- Psychological testing;

in combination with any of the following:

- Pregnant women as participants;
- Children aged five and under;
- Blood borne pathogens;
- Cell Gene Therapy, therapies targeting blood, blood barrier or cerebrospinal fluid;
- Any Product, Pharmaceutica or Medical Device that does not have the appropriate regulatory approval;
- Activities being undertaken by the lead investigator or any other member of the study team in a country outside of the UK;

- ☐ Yes
- ☐ Yes but I have discussed this question with my supervisor and have been advised to continue
- ☒ No

UGPGT4

Does your application contain any of the following:

- The use of equipment and/or substances that have not been used before, do not follow standard procedures and will be carried out in locations where a suitable and sufficient risk assessment is not already in place;
- Any hazardous activities, travel, substances, equipment, and/or locations that potentially pose a risk of ill health, injury or other harm (including stress) to participants, researchers or others;
- Any individuals or groups of people specifically at risk e.g. young persons, new and expectant mothers etc.;
- The need to obtain expert advice to ensure you have identified all reasonably foreseeable risks.

☐ Yes

☒ No

UG/PGT Application Information

Begin UGPGT Section

The following questions relate to the ethical concerns associated with your project.

☒ Begin section

UGPGT5 Is your project connected to another EthOS approved project?

☐ Yes

☒ No

UGPGT6 Please describe your research

Please make sure to discuss your aim, the activities you undertake, any special software/equipment you will use and the outcomes you expect.

Breast Cancer Ultrasound Classification Using Machine Learning

UGPGT Human Participants

UGPGT7 Does your research involve recruiting human participants?

This includes social media content (with or without the account holder's knowledge)

☐ Yes

☒ No

UGPGT Disclosures and Conditions

UGPGT10 Please summarise the main ethical issues arising from your project and say how they have been addressed

None

UGPGT11 By submitting this application you agree that you will do the following:

- If any of the details that you have declared in this application change then you will apply for an amendment;
- You will store all your project data (for which Manchester Met is Data Controller) in your student OneDrive account at Manchester Met;
- If your project uses personal data, you will store identifiable data separately from non-identifiable data and link them through a PIN (Participant Identification Number);
- You will destroy identifiable data according to the [University's retention and disposal schedule](#);
- You will not share identifiable data with any third-party without previously discussing this with dataprotection@mmu.ac.uk;
- You will maintain participant anonymity in publication unless the participant consented (in writing) to their identity being made public.

☒ I agree

Declarations

Y1 Do you have anything else you wish to declare or any documents you have not yet uploaded?

- ☐ Yes
- ☒ No

Signatures

Please be aware that the system will not process this application until all signatures (applicant/PI/supervisor) have been provided. Once all signatures have been provided then the application will be able to move to a 'submitted' status.

Please also be aware that 'submission' does not constitute approval - you will need to wait until an approval letter has been issued (via email) by the system. This email will arrive from 'donotreply@infonetica.net' - some email accounts will automatically sort this into your junk folder. This letter can also be downloaded in the system.

I confirm that all information in this application is accurate and true. I will not start this project until I have received an approval letter from the system.

☒ I confirm

Z2 Have you been instructed by your supervisor to request a second signature for this application?

- ☐ Yes
- ☒ No

Z3 Please click [here](#) to request your supervisor's signature. You will be asked to input their email address.

Your application will not be submitted until your supervisor has signed - it is your responsibility to ensure that they do this.

Signed: This form was signed by Dr Moi Hoon Yap (M.Yap@mmu.ac.uk) on 18/06/2026 11:18

Z6 By signing this application you are confirming that all details included in the form have been completed accurately and truthfully.

Signed: This form was signed by David Akerele (DAVID.D.AKERELE@stu.mmu.ac.uk) on 18/06/2026 10:56

Appendix E

Product Repository and Access

The digital product associated with this dissertation is available from the following repository:

[https://github.com/DavidAkerele/
breast-cancer-ultrasound-classification-ml](https://github.com/DavidAkerele/breast-cancer-ultrasound-classification-ml)

The repository contains the code for data audit, preprocessing, model training, evaluation, the API, and the browser dashboard. It also contains the stored results, executable notebook, documentation, automated tests, report source, and presentation source. This public repository is the stable product link for assessment.

The local demonstration address is <http://localhost:8000>. It works only after the repository has been downloaded and the service has been started.

From the repository root, the product can be checked with the following commands:

```
python -m pytest -q
uvicorn api:app --host 0.0.0.0 --port 8000
```

The implementation is a research demonstration and not a clinical service. Repository access should be tested after upload, and the submitted revision should remain available to the examiners until the end of the assessment period.